

Timescales of Earth and Temperature

Today we will work with a large printout (17 in $\times$ 11 in) of graphs¹ showing sequentially more recent histories of the Earth, with overlaid graphs of the estimated temperature² during that time. Let's call that the "*Timescales* printout." Also in this document are: (1) a list of "events" that have happened since the formation of the Solar System (the Sun and its planets); and (2) a graph of CO₂ over time.

For questions that require an answer, provide that on a separate sheet of paper (clearly indicating the question number and using brief but complete sentences). Other questions require that you mark-up the large printout. (I will have B&W versions to use as drafts prior to marking up the color one, if you would like).

We will work in small groups (~ 3) in our Discussion class but each student must complete their own version of the assignment. Whatever remains to finish after class will be your "Homework". The assignment is due (submission to the Blackboard assignment as a single pdf file) next Tuesday.

1. Getting familiar with the *Timescales* printout.

- (a) Briefly describe what is shown on the *Timescales* printout (What are the graphs plotting? What are the ranges on the axes and how do they relate? What are the calendar/clocks?).
- (b) Our Universe is about 13.8 Billion years old (that is, 1.38×10^{10} years). The calendars/clocks to the left of each graph are meant to express the time period of that row's axis as a fraction of a year, where one year represents the entire age of the Universe. The first row's calendar is shaded in to give this representation. Indicate the year fraction on the other four calendar/rows by shading or giving a time on the clock. (You should first figure out why I've only shown, e.g., December for the second row, and what the clocks are meant to represent.)
- (c) Consider the full range of time of the second row's axis. Indicate the time range of the second row *on the graph of the first row* (using a small arrow or a vertical line going through the first row's graph). Now indicate the third row's timescale on the second row's graph; and the fourth row's timescale on the third row's graph.

2. Plotting events on the "Timescales" printout

Using the provided list of events in the history of Earth (which includes geological, biological/life, human history, and "personal history" types of events), plot some events on the five timescale graphs. These are some rules/guidelines that you must follow:

- You are welcome use any of the events — and add your own events (from your History and Earth Science courses, and your life) — but **events on the list marked with a star (★)** should definitely be plotted.
- Find the "dates" of these events — specified as a certain number of "years ago" (ya) — by looking them up on your phone/computer.
- Mark each event with a dot/vertical-line at the correct "date." Label each event.

¹You will receive a copy of the graphs and this document in class, but always see Blackboard for the pdfs. Also we will have some B&W versions that you can use as drafts, if you like, before marking up the nice color one.

²See references to the original data sets below.

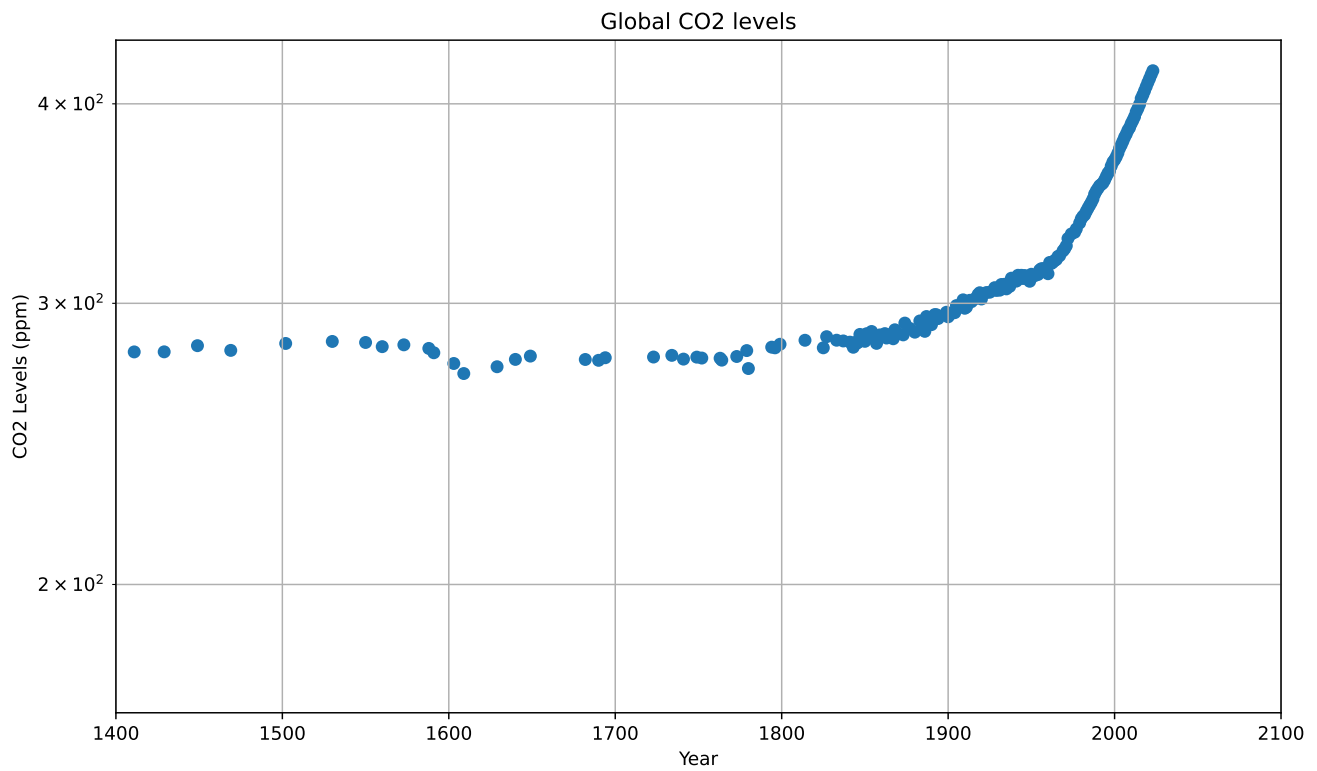
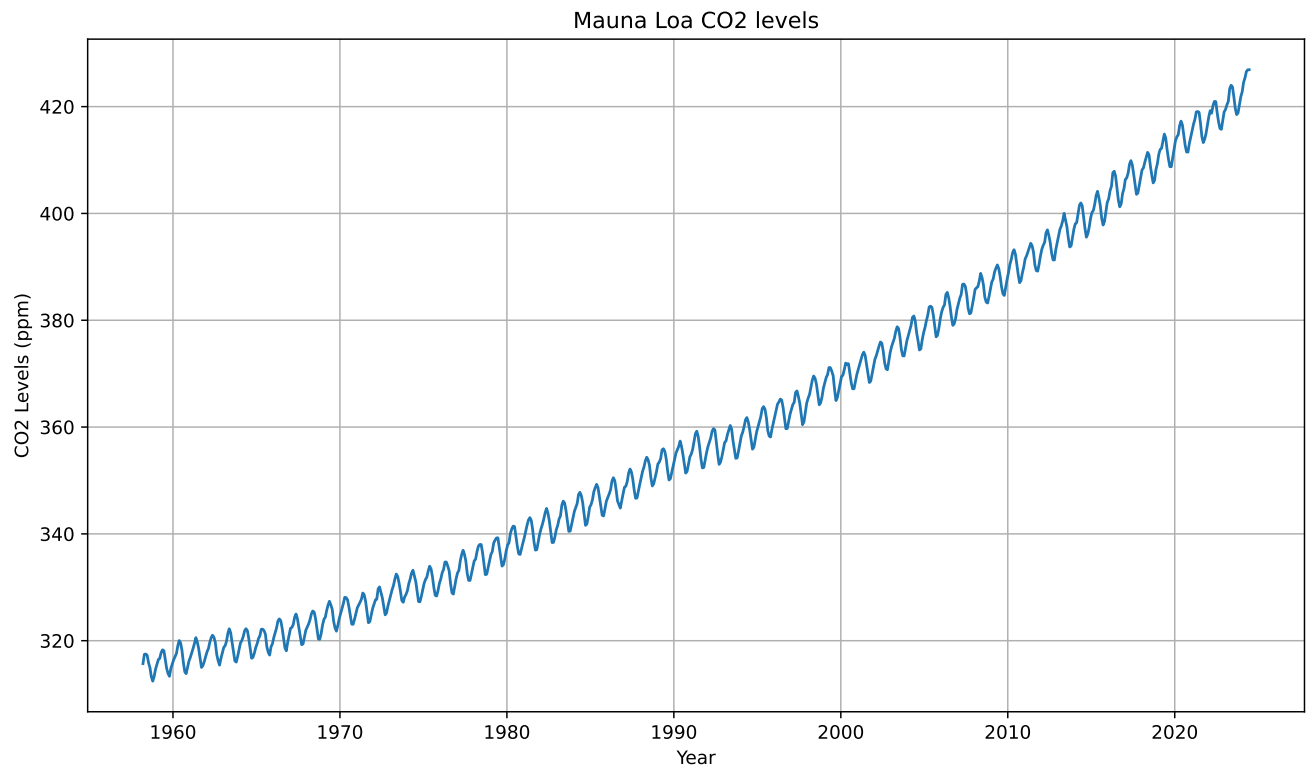
- For the first four rows, you should have a total of 20–40 events (5–10 per row), and **they must be (approximately) equally spaced across the axis** for that row, not bunched up in any one time period.
 - In the bottom row, indicate ~ 10 events, again **equally spaced across the entire axis**. Note that this graph has a *logarithmic*³ scale, so it actually spans the entire history of the universe (10^{10} ya is 10 Gya, 10 Billion years ago; and $10^{-1} = \frac{1}{10}$ ya is 5.2 weeks ago!).
3. **Atmospheric gases over time** A graph that estimates atmospheric carbon dioxide over time is provided.
- (a) On the CO₂ graphs, indicate the timespan of the first graph on the second graph, and the timespan of the second graph on the third graph, and the timespan of the third graph on the fourth graph.
 - (b) On the *Timescales* graph row three, put tick marks on the far right axis to create a CO₂ scale that will span the range of values in that time period.
 - (c) On the *Timescales* graph row three, indicate with stars the values of CO₂ at the time of at least three “ice age” and three “interglacial” periods.
 - (d) Look up online information about estimated atmospheric CO₂ concentrations from the period of the *Timescales* row two plot. Make an axis at right and plot a few points.

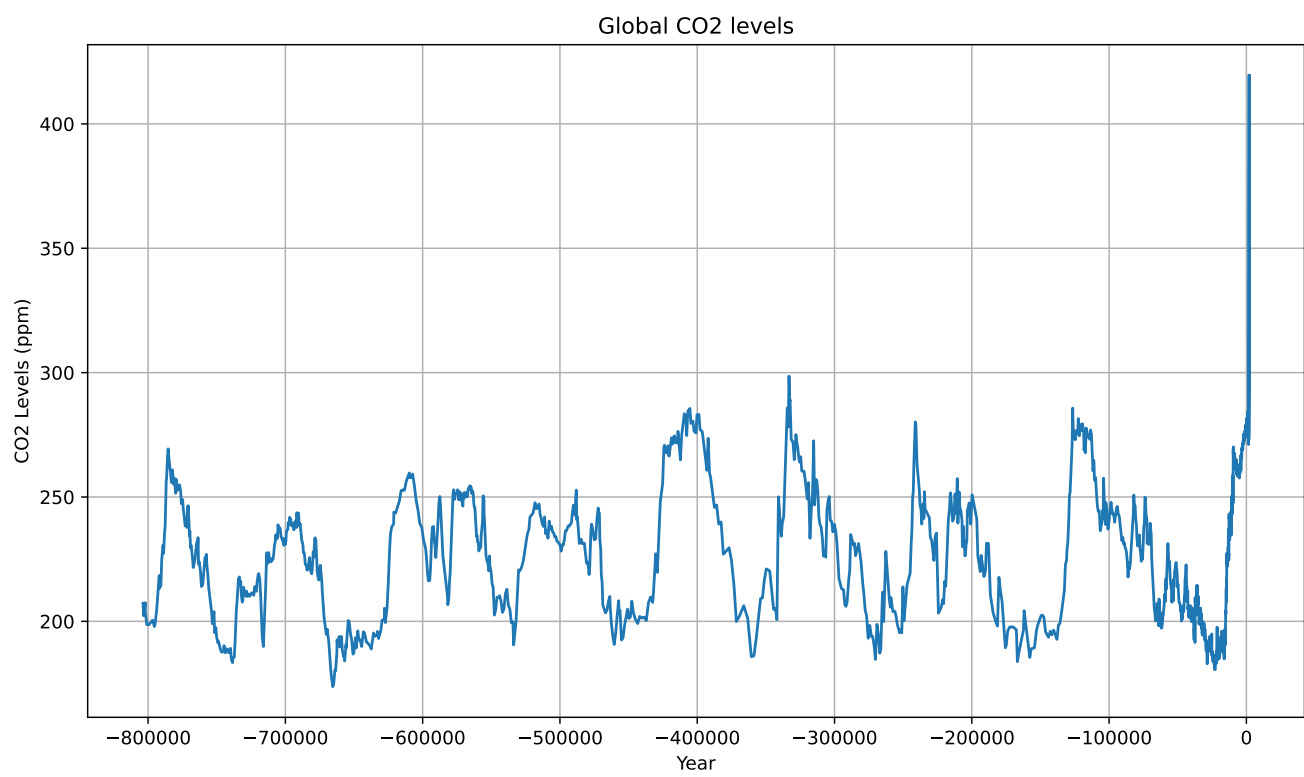
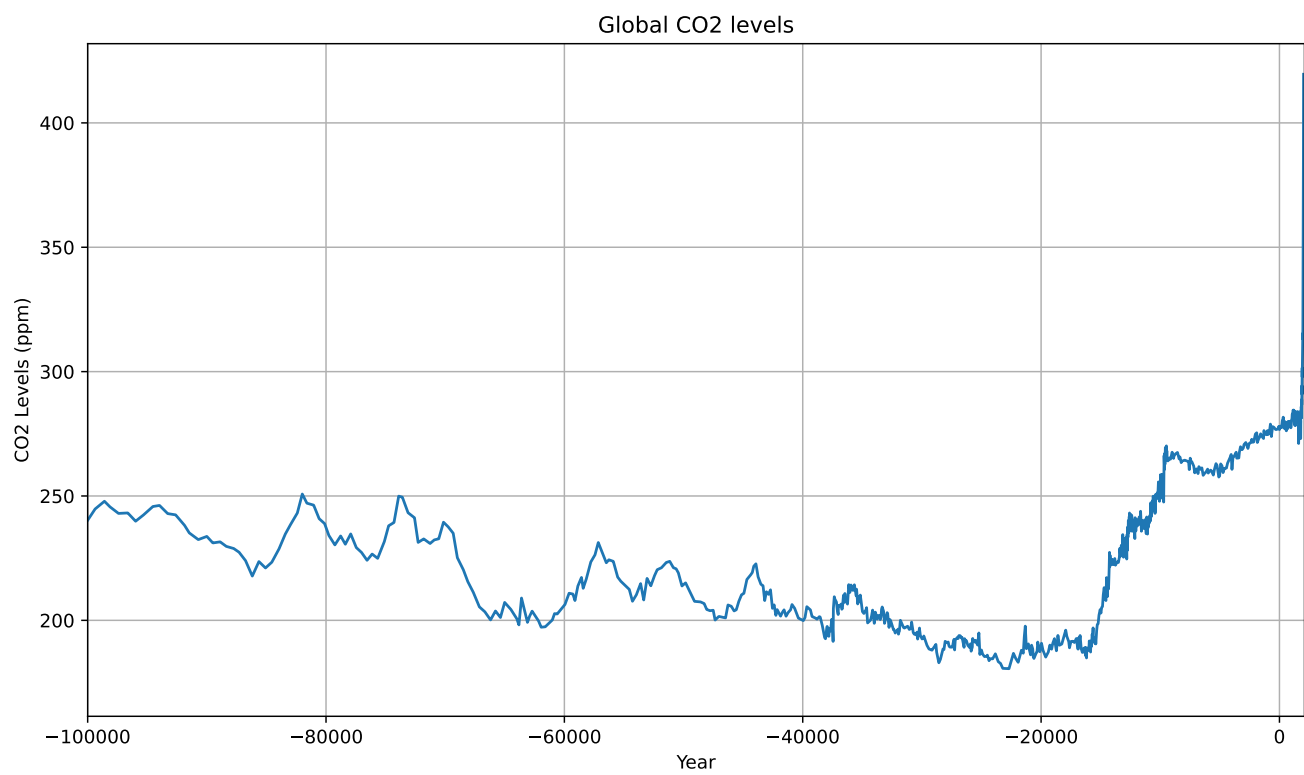
Data Sources:

- Data for the temperature/temperature-proxy plots was collected by Glen Fergus to create the very nice wikipedia graph “[All Paleotemps](#)”. As described on that page, the original published data sets for each row are:
 - 1: Veizer et al (Chem Geol 1999) as re-interpreted by Royer et al (Geo Today 2004).
 - 2: Zachos et al (Nature 2008) as interpreted by Hansen et al (Phil Trans R S A 2013); (with Veizer data).
 - 3: Lisiecki et al (PaleoOcean 2005); and EPICA Dome C (Antartica) ice core (Jouzel et al 2004); (with Zachos data).
 - 4: Marcott et al (Science 2013); NGRIP (Greenland) ice core (Anderson et al, Nature 2004); and Berkeley Earth data (10 y average of Global Temperature); (with Lisiecki and EPICA).
- Data for CO₂ is taken from the repository at [Our World In Data](#), which uses the [NOAA Global Monitoring Laboratory — Trends in Atmospheric Carbon Dioxide \(2025\)](#) dataset.

³Starting at the right tick mark, $10^{-1} = 1/10$ ya, the next tick mark to the left is 10 times longer ago ($10^0 = 1$ ya), and then the next is ten times longer ($10^1 = 10$ ya), and then 10 times longer ($10^2 = 100$ ya).

Atmospheric CO₂ Data (from recent measurements and ice core bubbles)





Example “events” in history (feel free to add your own!)

Cosmological / Planetary / Geological

The Big Bang
Formation of our Milky Way galaxy
Formation of our Solar System
Formation of our Earth
Formation of the Moon
“End of late heavy bombardment”
Departure of today’s light from Alpha Centauri
Formation of oceans
Supercontinents (Rodinia, Pannotia, Pangea)
Last glacial maximum (ice age)
Last “interglacial period”
First ice age
First “interglacial”
Formation of the Great Lakes
Formation of the Appalachian Mountains
Formation of the Rocky Mountains
Bering land bridge submerged

Humanity (and cousins)

First appearance of Genus “Homo”
Beginning and end of: Homo Habilis, Homo Erectus, Homo Neanderthalis, Homo Sapiens
First humans in: Europe, Asia, North America
Ancient inventions: wheel, ...?
Ancient Societies: Greek, Egypt, Sumer, Shang...?
Roman Empire
Founding of modern nations/societies: England, China, USA, India
Modern Inventions: steam engine, airplane, computer,
Energy: First use of oil, coal; Solar power,

Biology / Life

First single-celled organisms (Prokaryotes)
First photosynthesis
Appearance of atmospheric oxygen
First Eukaryotes (cells with nuclei and sex reprod)
First multi-cellular life
First appearance of: plants, animals, fish, amphibians, reptiles, mammals, dinosaurs, primates
“Cambrian Explosion”
End of Dinosaurs
Last common ancestor pairs: Humans-Chimps, Humans-Monkeys, Humans-Whales, Humans-Sharks, Horses-Whales, ...

You and Yours

Your birth
Birth of: parents, grandparents, ...
High school graduation, your parents
Founding/incorporation of hometown
House built
Favorite sport team’s founding

Contemporary Climate Change: How do we know what we know?

1. Detection of Climate Change

- (a) In the reading and in lab this week we discussed the “global average temperature anomaly” (GATA). Briefly explain how this quantity is defined and how it is calculated, given measurements of temperature from thermometers at various locations on Earth.
- (b) Pointing to specific places on the graph of GATA¹ 1850–present (see figures on next pages), comment on: (i) how the methodological *uncertainty* in the estimate of the GATA has changed over time (consider for example the comparison of the values from independent groups; try to give quantitative estimates) and, (ii) why the uncertainty might have changed in this way.
- (c) Pointing to specific places on the graph, comment on: (i) the natural variability in the *value* of the Earth’s temperature (Can you give a quantitative estimate for the value of this variability in °C?), and (ii) which features of the data cannot be explained by “natural variability.”
- (d) Consider now the longer-term temperature graph — including both human thermometer measurements since 1850, and geological “temperature proxy” measurements since 25,000 years ago — comment again on the “natural variability” of Earth’s temperature, and specifically how we should interpret the change in GATA since 1970.

2. **Attribution of Climate Change** Scientists in the “working group” for *the physical scientific basis of climate change*² within the Intergovernmental Panel on Climate Change (IPCC), have use the terminology “Detection and attribution” of climate change³. Problem (1.) addressed the question of detection: Has the climate changed in a statistical sense, i.e., outside of the bounds of instrumental and methodological uncertainty, and natural variation? (I think your answer to that question is “yes.”) Here we consider the *attribution* question: To what can we attribute these changes; what is their cause? This is often stated more specifically as: Can the detected increase in GATA be confidently attributed to human greenhouse gas production?

- (a) Consider the information presented in the lecture slides (see “Class Slides” on Blackboard, possibly supplemented by Nadir Jeevanjee’s original talk). State and briefly explain 2–3 pieces of evidence that support the argument that our warming climate can indeed be attributed to greenhouse gas (primarily CO₂) production.
- (b) Consider the 2008 Geophysical Research Letters paper by Lean et. al.⁴, whose abstract and first two figures are shown on a later page. (i) What do they mean by a “robust multivariate analysis”? (ii) What is the connection between the data shown in Figure 1 and the overplotted green line on the temperature graph of Figure 2? (iii) What is the overall conclusion of this analysis?

¹In the figure it is called the “global mean temperature difference.”

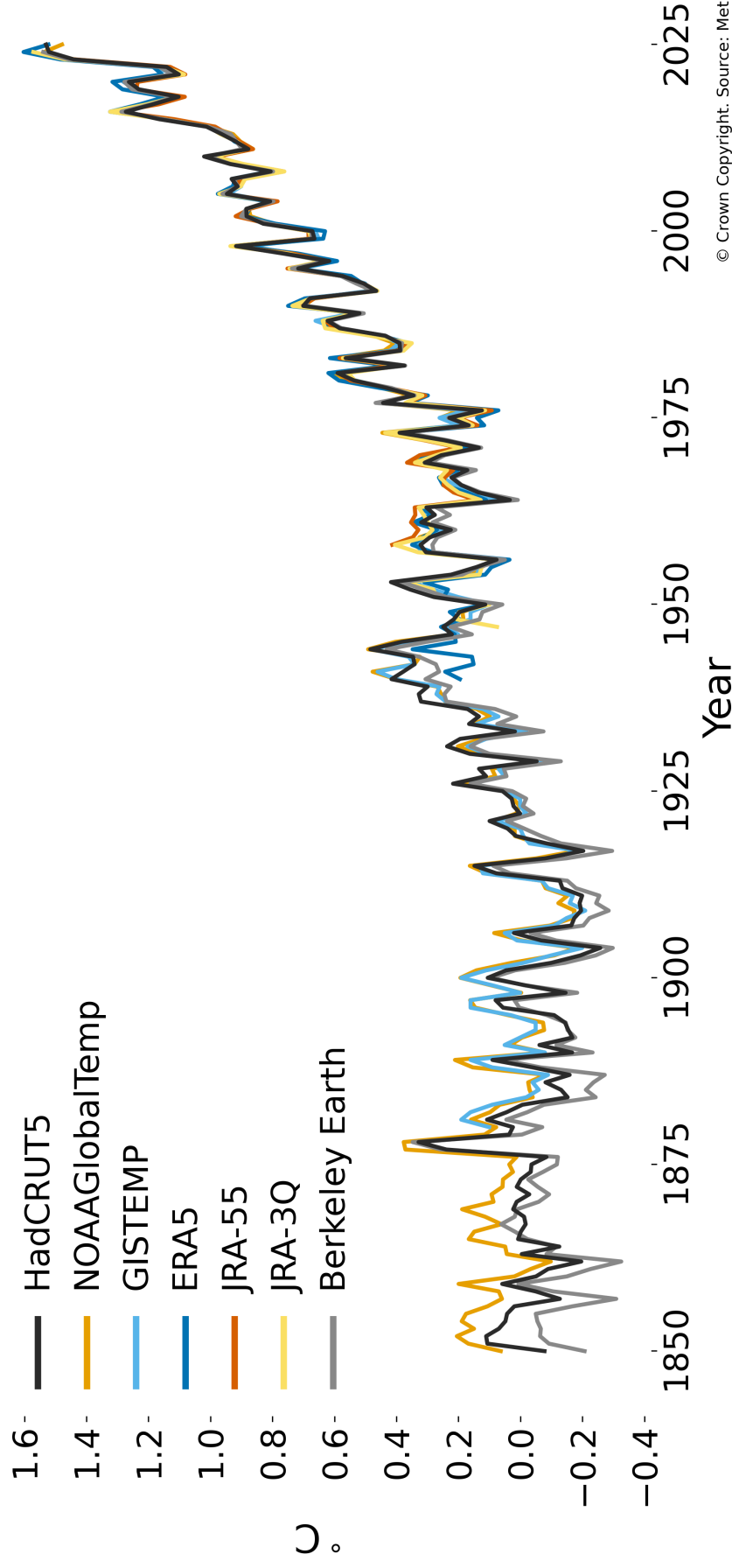
²There are two other working groups in the IPCC: WG2 assesses the vulnerability of socioeconomic and natural systems to the effects of climate change, and WG3 assesses options for mitigating climate change.

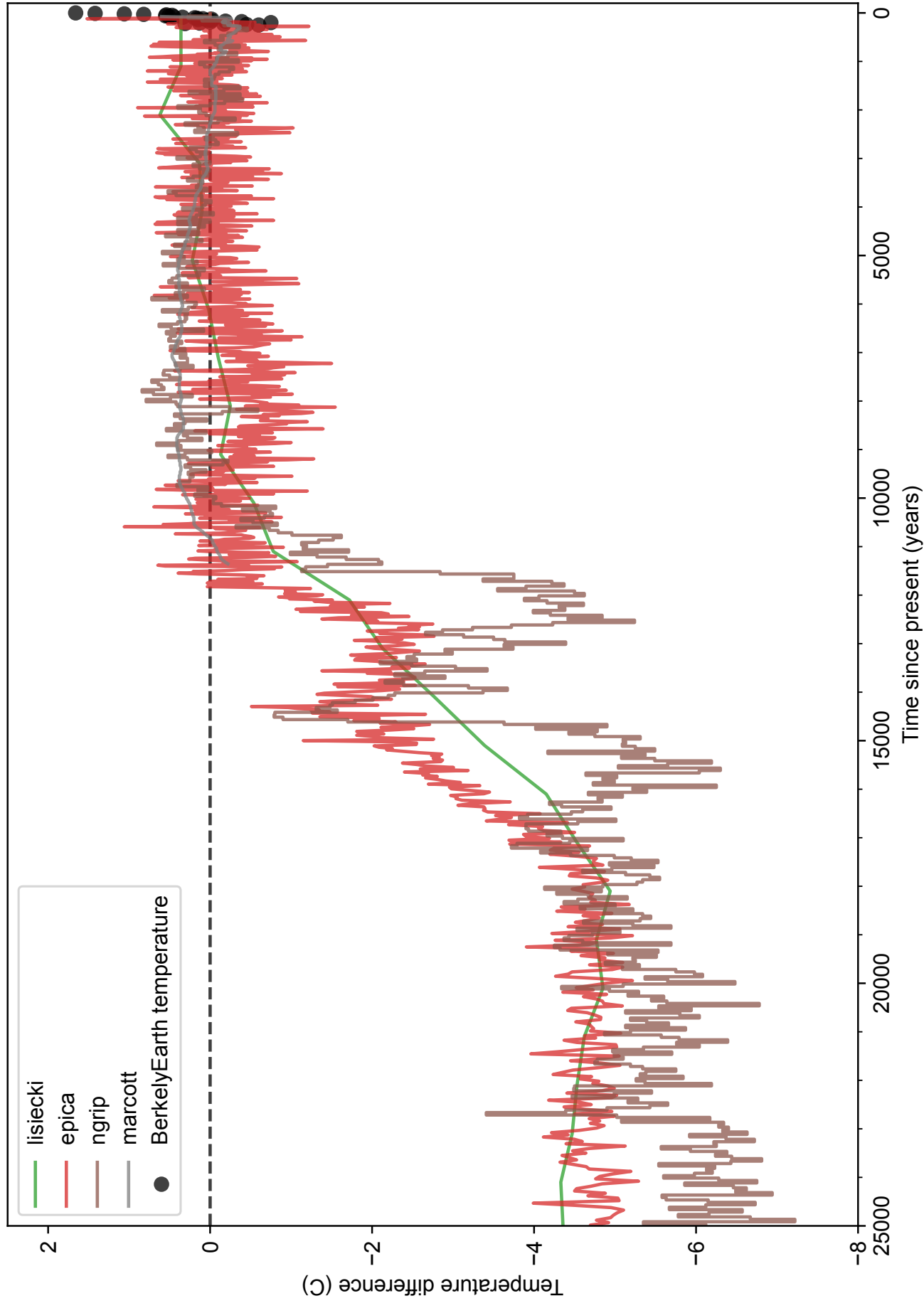
³See a complete definition and explanation of the terminology in the 2007 WG1 report [here](#), as well as the [AR5 \(2018\)](#) report.

⁴From Lean et. al. *How natural and anthropogenic influences alter global and regional surface temperatures: 1889 to 2006* ([Geophys Res Lett 2008](#))

- (c) There is another interesting piece of evidence that allows us to attribute warming to greenhouse gas emissions, which is often called the “human fingerprint” of temperature change. (i) Read the abstract of Santer et al (PNAS 2023) linked on Blackboard and describe what the effect is. (ii) Using Figure 1 of their paper, discuss how they have observed the effect.

Global mean temperature difference from 1850-1900 (° C)





How natural and anthropogenic influences alter global and regional surface temperatures: 1889 to 2006

Judith L. Lean  David H. Rind

First published: 16 September 2008 | <https://doi.org/10.1029/2008GL034864> | Citations: 280

Abstract

[1] To distinguish between simultaneous natural and anthropogenic impacts on surface temperature, regionally as well as globally, we perform a robust multivariate analysis using the best available estimates of each together with the observed surface temperature record from 1889 to 2006. The results enable us to compare, for the first time from observations, the geographical distributions of responses to individual influences consistent with their global impacts. We find a response to solar forcing quite different from that reported in several papers published recently in this journal, and zonally averaged responses to both natural and anthropogenic forcings that differ distinctly from those indicated by the Intergovernmental Panel on Climate Change, whose conclusions depended on model simulations. Anthropogenic warming estimated directly from the historical observations is more pronounced between 45°S and 50°N than at higher latitudes whereas the model-simulated trends have minimum values in the tropics and increase steadily from 30 to 70°N.

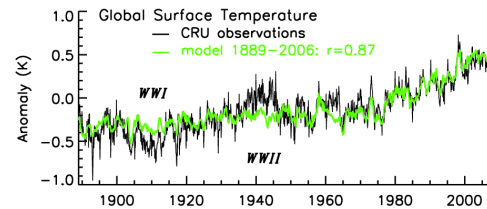


Figure 1. Compared with the CRU monthly mean global temperature time series (hadcrut3vcgl) is an empirical model obtained from multiple regression for the period from 1889 to 2006, inclusive. The value of r is the correlation coefficient for the global temperature observations and empirical model. Largest differences occur at the times of the two World Wars when observations were sparse.

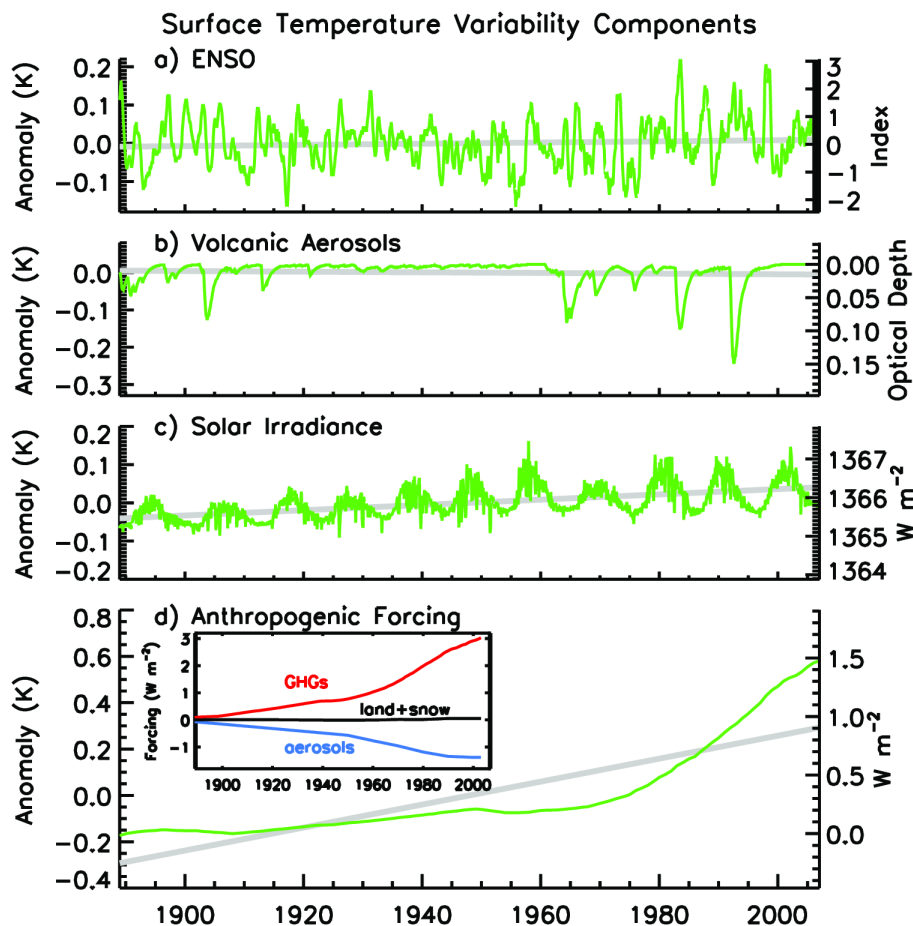


Figure 2. Reconstructions of the contributions to monthly mean global surface temperatures by individual natural and anthropogenic influences (at appropriate lags) are shown. The right hand ordinates give the native scales of each influence and the left hand ordinates give the corresponding temperature change determined from the multiple regression analysis. The grey lines are trends for the whole interval. The inset in Figure 2d shows the individual greenhouse gases, tropospheric aerosols and the land surface plus snow albedo components that combine to give the net anthropogenic forcing.

Global Mean Temperature Changes in Model and Observed SSU and MSU Data

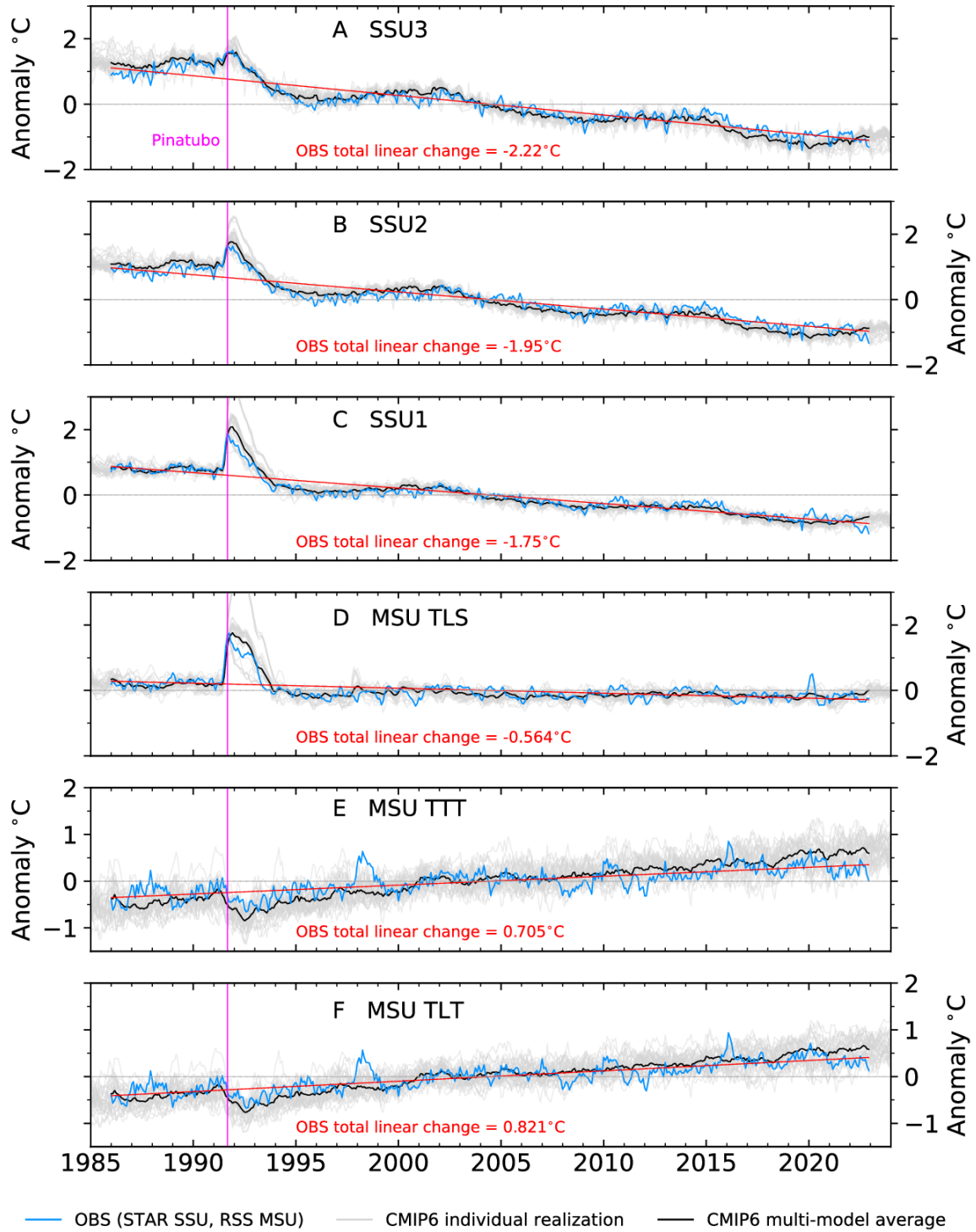


Fig. 1. Observed and simulated changes in global-mean monthly mean temperature in six atmospheric layers. Results are temperatures from channels 3, 2, and 1 of the Stratospheric Sounding Unit (SSU; panels A–C) (27), lower stratospheric temperature from the Microwave Sounding Unit (MSU TLS; panel D), MSU total tropospheric temperature (TTT; panel E), and MSU lower tropospheric temperature (TLT; panel F) (25). The peaks of the weighting functions for these six layers are at ca. 45, 38, 30, 19, 5.6, and 3.1 km above the Earth's surface (respectively). Results are anomalies relative to climatological monthly means over 1986 to 2022. Model simulations are from nine different CMIP6 models and a total of 32 realizations of historical climate change (*Methods* and *SI Appendix*).

Energy and Equilibrium

1. **Water Flow Diagram and Analysis:** Our basic law of energy is its *conservation*: although it might change its form, or transfer from one object to another, it is neither destroyed or created. In the next question, we will try to sketch that idea by diagramming systems of objects: we will account for changes in the energies of individual object by the transfers of energy between those objects.

Water is similar to energy because it is also (approximately¹) conserved. Let's study the example of our "leaky bucket" experiment from lab this week. The water that starts in the pitcher is never "lost," although, at any later moment, it might be in another component of the experimental setup (or on your table).

- (a) Describe generally in words how water flows through the leaky bucket system. At some random moment during the experiment, where is all of the water that started in the pitcher? (You might want to start part (b) in parallel to answering this part, particularly the picture, to remind yourself).
- (b) Let's construct a "Water Flow Diagram" for the leaky bucket system. Start with a blank page of paper.
 - At the top of the page, draw the experimental setup from the lab. Label the objects, including: pitcher, "leaky bucket" (100 mL grad cyl w/ hole), catch-tray, pump,
 - Just below the picture, title the system and state the brief interval of time we will analyze. For this first diagram, take the interval of time to be: *after you increased the spigot flow rate but between equilibrium water levels (water level is rising)*.
 - In the bottom half of the page, draw a square box for each object in the system. Split each box into four quadrants and label them as:
 - Top-left quadrant: "Name" of object
 - Top-right: blank (used in Question 2, when we change water to energy)
 - Bottom-left: blank
 - Bottom-right: "Total"
 - In each object box, give the name, and then indicate in the "Total" box whether the total water in that object is increasing/decreasing/neither during the chosen time interval (using up/down/horizontal arrows).
 - Now draw an arrow *between any two boxes* for which there is a *flow* of water from one to the other. It is possible that water flows out of an object to multiple other objects, or that an object receives water from multiple objects.
- (c) Check to make sure that any object with a changing "Total" water inside, has arrows in/out of it that *explain* that change. Discuss this briefly for each object.
- (d) Let's make part (c) at least qualitatively quantitative (ha). Assign an arbitrary number to one of your arrows indicating a flow (perhaps the water leaving the pitcher spout, maybe "100"). Now move through your schematic system and account for that number

¹It is not quite as strict a conservation law here. If we boil water it becomes steam; if we freeze it it becomes ice. So to "conserve" liquid water we restrict ourselves to room temperature conditions and ignore evaporation. More dramatically, even if we were to include all phases of water, the molecules of H₂O can be lost, in the process of "water-splitting"/electrolysis: $2\text{H}_2\text{O} (l) \rightarrow \text{H}_2 (g) + 2\text{O}_2 (g)$.

throughout the rest of the system. Assign numbers to all flow arrows and to all “Total” boxes within objects. Make sure that any “Total” change number is exactly equal to the amount flowing into that object, minus the amount flowing out of that object.

- (e) Now construct a second “water flow diagram” but choose the time interval to be “once the leaky bucket water level has come to equilibrium.” Complete all of the steps in (b)–(d) (but you do not need to re-draw the picture of the experimental setup).

2. **Energy Flow Diagrams:** We will now construct “energy flow diagrams” to show energy flow between the objects in a system causing “total” energy changes for the individual objects. The energy flow diagram will have the same basic structure as the water flow diagrams of the previous question — i.e., picture of system, stated choice of brief time interval, schematic with boxes for objects and arrows for flows — but with a few additions:

- Within each object box, label the “Name” and “Total” as in Question (1). Label the other quadrants now as:
 - Top-right quadrant: “External”
 - Bottom-left quadrant: “Internal”
- Within the External and Internal quadrants you will list *energy types* (see info sheet on last page), paired with with up/down/horiz arrows indicating their change over the chosen time interval. There could be multiple types of energy in each quadrant.
- As before, within the “Total” quadrant show the total energy change using a single up/down/horiz arrow. This must again be consistent the net flows in/out of the object. But now it must also agree with the “sum” of the External and Internal changes.
- Next to each “flow” arrow, specify the *energy transfer mechanism* (see info sheet).
- As before, make things “qualitatively quantitative” with numbers. But now also include numbers next to each energy type listed within the External/Internal quadrants, and ensure that their sum matches the value in the Total quadrant. The Total change of energy in an object is equal to the net flow of energy in/out of the object, but it is also equal to (explained by) the sum of the changes in the internal/external energy types.

Make an energy flow diagram for each of the following situations. We’ll do part (a) together.

- (a) A person (with spiky shoes) drags a box across a frictionless surface (e.g., ice). They pull with a constant force. Let the time interval be “shortly after they start pulling.”
- (b) Same as previous, but now there is friction between the box and the floor. The box moves with a constant speed².
- (c) Heating a pot of boiling water on the stove. Do it twice: once for “shortly after you put the pot on the stove” and once for “once the water is fully boiling”.
- (d) An ice cube melting on a picnic table in the sun.
- (e) A situation of your choice that includes an object in energy equilibrium.
- (f) A situation of your choice that includes an object not in energy equilibrium.

²Because the box moves with a constant speed, the “Work-Energy Theorem” tells us that there must be two works that change the energy of the box: pulling of the person (positive work) and friction from the floor (negative work). These two works sum to make $W_{\text{tot}} = 0$, so $\Delta K = W_{\text{tot}} = 0$.

3. **Leaky Bucket Theory and Data Analysis (BONUS for F2025):** In the “leaky bucket” lab experiment this week, you should find that your final plot — flow rate (mL/s) vs height (cm) — does not seem to be a straight-line (linear) dependence. Let’s figure out why.

- The basic definition of *equilibrium* is that the flow rate into the system is equal to the flow rate out of system. If one of those two rates depends on a *parameter* in the system (something that describes the *state* of the system), then equilibrium for the system will imply an *equilibrium value* for that parameter. As an equation, where F means *flow rate*, and x is the parameter:

$$\text{“equilibrium”} \rightarrow F_{\text{in}} = F_{\text{out}}(x) \rightarrow \text{solution for equilibrium value: } x_{\text{eq}}$$

where the parentheses on the right-hand side of the flow-rate equation means that the outflow rate *depends on* the parameter x .

- In our “leaky bucket” lab, the graduated cylinder (GC) is the “system.” Equilibrium for the GC means that rate of inflowing water (from the pitcher) is equal to the rate of outflowing water (through the hole in the GC). The height of water in the GC (above the hole) is the “parameter” of the system on which one of these rates must depend. Because the inflow rate depends only on how much you turn the spigot (and not on anything about the GC system), it must be that the outflow rate depends on the height parameter³. In summary of the experiment: you set an inflow rate by adjusting the spigot, you wait until you observe the height to come to its equilibrium level, which is achieved because the outflow rate through the hole matches the inflow rate.

By answering the following questions, we will derive the explicit dependence of outflow on height and be able to analyze the graphed lab data.

- In the experiment, how do you *see* that the outflow rate from the GC is changing? The hole is always the same size, so how does the system seem to adjust to let more water out when the inflow rate from the pitcher increases?
- The change that you see in outflow can be accounted for in a (pretty) simple law of fluid dynamics, called *Bernoulli’s Principle*:

$$P_A + \rho gh_A + \frac{1}{2}\rho v_A^2 = P_B + \rho gh_B + \frac{1}{2}\rho v_B^2.$$

This says that the sum of the pressure of a fluid, P , plus its gravitational energy density⁴, ρgh , plus its kinetic energy density, $\frac{1}{2}\rho v^2$, is equal at any two points (A and B) along a streamline of a fluid. To apply this to our lab, where the “fluid” is the water in the GC, we pick our points A and B to be at the top of water column and where the water shoots out of the hole, respectively. Because both of these points are exposed to the

³It is possible that neither rate depends on a parameter, in which case there is no “equilibrium value” for a parameter because no equilibrium is achieved. But that does not match observations for the leaky bucket experiment. If the outflow rate of the GC did *not* depend on water height then, once the inflow rate from the pitcher exceeded the (fixed) outflow rate from the hole, the graduated cylinder would completely fill up and overflow. You would either see a graduated cylinder full to the hole (inflow less than outflow), or a graduated cylinder overflowing (inflow greater than outflow), but nothing in between.

⁴ ρ is the *mass density of the fluid* (1 g/mL = 1000 kg/m³ for water); and $g = 10 \text{ m/s}^2$ is the so-called *acceleration due to gravity* constant.

atmosphere, the fluid pressures are the same: $P_A = P_B = P_{\text{atm}}$. Because, at equilibrium, the water level is not changing, what is v_A , at the top of the water column? If we say that the height of the hole is our zero value, $h_B = 0$, then $h_A = h$ is simply our “height above the hole.” Make all of these replacements in the above equation and you should arrive at an equation that relates v_B , the speed of water shooting out of the hole, to h_B , the height of the water level above the hole. We can just call these v and h from now on.

- (c) Now we must relate the speed of the water shooting out the hole, v , to the flow rate, F . Flow rate is measured in volume per second (e.g., mL/s or cm³/s), while speed is measured in distance over time (e.g., cm/s). From their units, what type of thing should you multiply speed by to get something in units of flow rate? Write this down as an equation.
- (d) It turns out the equation you “derived” in part (d) is correct⁵. Combine your two equations — in parts (b) and (c) — to get an expression that relates F to h .
- (e) What should your final expression look like as a graph, in particular as a function of F vs h (with F plotted on the vertical axis and h on the horizontal). Does the data plotted in the lab experiment agree with that general behavior?
- (f) Although your final expression is not a *linear* relationship (the flow rate is not just proportional to the height), you can make your data fit a linear relationship by adjusting either the F data or the h data (by, for example squaring one of them). Try this out and draw a line of best fit through the data in this case. What is the slope of your data fit and what should it be? [The diameter of the hole in your GC was 11/64 in = 0.4365 cm, and the area of a circle is πR^2 .]

⁵You can derive it more carefully by considering a small cylinder of water that shoots out the hole in a small time interval Δt . The length of that cylinder will be $\Delta x = v \times \Delta t$ (a re-arrangement of “speed is distance over time”). And the volume of that cylinder will be $V = \Delta x \times A$, where A is the circular area of the hole. Therefore the volume rate of flow is: $F = V/\Delta t = (\Delta x A)/\Delta t = \frac{\Delta x}{\Delta t} A = vA$.

Energy Information Sheet

Energy Types (associated to an object)

“External” Energy (mechanical energy associated object’s motion or interaction with fields)

- **Kinetic Energy:** Energy of motion, depending on mass and speed as $K = \frac{1}{2}mv^2$.
- **Potential Energy:** Stored energy. Two common examples: compressing a spring ($V_S = \frac{1}{2}k\Delta\ell^2$); and raising an object in the gravitational field ($V_g = mg\Delta h$)

“Internal” Energy (energy within the defined boundaries of the object)

- **Thermal kinetic energy:** kinetic energy of free atoms/molecules (in a fluid)
- **Thermal potential energy:** potential energy of atomic/molecular vibrations (in solid)
- **Chemical energy:** stored energy in the bonds of molecules (sugar from food, ATP)
- **Internal Kinetic energy:** mechanical energy associated to motion of internal parts
- **Internal Potential energy:** mechanical energy in internal springs or rubber bands

Energy Transfer Mechanisms (between objects)

- **Work:** $W = F \times \Delta x$, mechanical energy transfer to/from an object by a force acting on it. Forces directed with motion yield positive work (kinetic energy gained⁶ by object); forces directed opposite to motion yield negative work (kinetic energy lost by object).
- **Radiation:** light/warmth, non-contact transfer.
- **Conduction:** energy transfer between two objects in contact with different temperatures (energy passes from the hot object to the cold object).
- **Convection/Mixing:** bulk motion of material within an object carries energy from one place to another (potentially followed by a radiative/conductive transfer)
- **Electrical Work:** Energy delivered by an electrical circuit.

⁶Change in kinetic energy due to work results from the “Work Kinetic Energy Theorem”: $W_{\text{tot}} = \Delta K$.

Energy Flow Diagrams (Basic Elements/Steps)

- Use a whole sheet of paper.
- At the top of the page, draw a **picture of the system** under consideration.
- Just below the picture, state the **name of the system** and the brief **time interval**⁷ under consideration.
- In the bottom half of the page, draw **schematic for the system**, including:
 - a **box for the energy changes of each object**
 - **arrows for energy transfers between objects**
 - **numbers to indicate conservation** (within boxes and on arrows)
- Details of the schematic:
 - Each object box has four sections: the “Name”; and the “External,” “Internal,” and “Total” energy changes.
 - “External” and “Internal” sections should have a list of the energy types associated to that object⁸
 - In the internal and external quadrants, state an energy type and indicate if it increases/decreases using an up/down arrow (or a horizontal bar for “no change”).

UNFINISHED (above and below!)

Analogy to water and water flow

- Basic idea is that liquid water (in a room temperature situation, ignoring evaporation) is conserved. It flows from one object to another, but the net flow into any object must equal the change in the amount of water that object contains.
- **Conduction** (transfer of energy from a hot object to a cold object in contact): When two wet objects are in contact (e.g., two sponges, one wet one mostly dry), water will flow from the wetter object to the dryer object (and never in reverse). The flow stops when the wetness of the two objects is equal. Wetness is the analog of temperature.
- **Radiation** (transfer of energy from one object to another by light/photons): Spray bottle sends small particles of water (if the analogy held, then the color of water and the amount per second out of the spray bottle would depend on the temperature (wetness) of source).

⁷These schematic “flow” diagrams describe the *transfer* of energy between objects of a system which causes *changes* in the total energy associated to each object. To make these abstract ideas more concrete — like the analogy of water flow, where 2 mL poured into a cup over five seconds increases the water inside by 2 mL during that 5 s time interval — we specify a brief time interval over which the transfers and changes are taking place. Over that time interval, Δt , we can then imagine the associated energy changes — ΔE_{tot} , ΔE_{ext} , ΔE_{int} — *of an object* and the energy transfers — mechanical work done, radiation flux of photons, heat transfer in conduction — *between objects*.

⁸We will only list the energy types that are *changing during our chosen time interval*. A box being dragged across a frictionless surface has internal chemical energy in the bonds between its atoms, but these are not changing during the time interval so we do not include them. On the other hand, the person dragging the box is able to exert a force through their muscles (and thus transmit energy to the box through work) because their cells are constantly breaking the bonds of ATP molecules to release stored chemical energy. So we *do* including internal chemical energy for the person, because it is decreasing during the chosen time interval.

Electromagnetic Radiation and Blackbody Radiation

Recall from lecture and lab, our basic equations concerning blackbody radiation:

$$\lambda_{\text{peak}} = \frac{2900 \mu\text{m} \cdot \text{K}}{T} \quad \text{Wien's Law, wavelength of peak emission}$$

$$F = \left(5.7 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T^4 \quad \text{Stefan-Boltzmann Law (total flux)}$$

$$F_{\lambda} = \frac{F}{1.5 \times \lambda_{\text{peak}}} \quad \text{peak value of "spectral flux"}$$

The “flux” is the power (energy per time) emitted per area (so, units of $\frac{\text{J}}{\text{s} \cdot \text{m}^2}$), and the “spectral flux” is what we graph versus wavelength. You can see¹ that spectral flux is flux per wavelength, so that the area under the F_{λ} versus λ graph — an approximate “triangle” of wavelength-range times spectral flux (base times height) — gives us the total flux.

Two other important equations, related to electromagnetic radiation (light) are:

$$c = \lambda f \quad \text{relation between wavelength and frequency}$$

$$E = hf = \frac{hc}{\lambda} \quad \text{energy per photon of light}$$

where $c = 3.0 \times 10^8 \text{ m/s}$ is the speed of light (literally, the speed at which light travels in a vacuum²). The second equation is a result from quantum mechanics which says that light is “quantized” into little packets of energy, called “photons”. The energy carried by each photon depends only on its wavelength/color³.

1. **Graphing Blackbody Radiation** You have been provided “log-log” graph paper, that allows us the plot quantities over a wide range of values, for each axis (recall that we fit the entire history of the universe on one plot, in the “Timescales” discussion in the first week).

Using the “rules of thumb” given in Table 1, plot approximate blackbody curves (spectral flux vs. wavelength) for these three objects⁴, all on the same plot:

- (i) The Sun,
- (ii) A Human,
- (iii) A vat of liquid nitrogen.

For each object, look up the temperature in Celsius and convert to Kelvin ($T_{\text{(K)}} = T_{\text{(°C)}} + 273$). Before drawing anything, find the values of peak spectral flux and peak wavelength for each (see “rules” below) and make a little table for yourself. These values should guide how you set the range on the graph paper (what is the minimum and maximum value on each axis). Then describe in words, what you see on the graph. How do the three objects compare in

¹You’ve noticed that I’m not showing you the actual equation for spectral flux, but only an expression for its peak value. We will do approximate drawings of the graphs $F_{\lambda}(\lambda)$ vs λ , using some rules of thumb, in Question 1.

²It takes light 8 minutes to reach us from the Sun. How far away is the sun? Use speed equals distance over time!

³I will often use “color” as a shorthand way of saying the wavelength or frequency of light. But our actual perception of color is quite different. Our eyes have three types of “cone” cells, each of which “collect and count” photons in their own little segment of the visible light wavelength spectrum: red, green, and blue. The brain then “adds” the contributions of the RGB buckets together to give us what we perceive as a color.

⁴Recall that the phrase “ y vs. x ” means that y is on the vertical axis and x is on the horizontal axis.

- Knowing the temperature of a particular blackbody object, you can calculate the peak wavelength using Wien’s Law and the peak spectral flux value using the “ $F/(1.5\lambda)$ ” equation. Plot these two values as one point on the F_λ vs. λ axes.
- Estimate the two wavelengths (one smaller than λ_{peak} , one larger) where the spectral flux graph falls to 10% of its maximum value (dividing by a factor of 10 means one large division down on a logarithmic axis) using the following rule of thumb: $\lambda_{\text{small}} = \lambda_{\text{peak}}/2$ and $\lambda_{\text{large}} = 3\lambda_{\text{peak}}$. Plot two more points on the F_λ vs. λ axes.
- Make a smooth triangular “hat” connecting these three points, the BB curve peak.
- On log-log graph paper, the tail of the BB spectral flux graph toward large wavelengths is exactly a straight line (with a slope of negative 4). You can draw the tail toward small wavelengths also approximately straight, but it actually drops off more quickly (so you could sketch a light straight line, but then draw something under it (that passes through λ_{peak} and λ_{small}).

Table 1: Rules of thumb for drawing blackbody curves

power output (if you had equal-sized 1 m² sun, person and N₂(ℓ) ball)? In what wavelength bands (visible, infrared, UV, . . .) are the objects emitting most of their power?

2. **Solar incident radiation and what is received by Earth’s surface:** We are interested (Question 3, below) in understanding how the incoming solar radiation determines the temperature of the Earth. The first step in this process⁵ is to determine how much energy the Sun provides to the Earth, i.e., how much average power “lands” per meter squared on the surface of the Earth?

- Energy leaving the Sun:** The flux of a blackbody is the energy (e.g., in Joules) released per second from every square meter of its surface area. Write down an equation for the total power released by a blackbody. Remembering that the surface of a sphere is given by $A_{\text{sphere}} = 4\pi R^2$, and using the astronomical information in Table 2, calculate the total power (in watts $W = \frac{J}{s}$) released by the Sun.
- Energy reaching the Earth from the sun:** The total energy leaving the Sun in one second is distributed over large spherical shell in all directions. Only part of that sphere intersects the Earth. Find the energy per meter that is caught by the Earth, using your answer to (a) divided by the total area of a sphere of radius $d_{\odot \rightarrow \oplus}$. This is called the *solar constant*.
- Spreading the sun’s light over the earth’s surface:** The total solar energy hitting the Earth is spread over its whole spherical surface. Show that your answer to (c) should simply be divided by 4 to achieve this.
- Albedo:** The “albedo” — denoted by the Greek letter alpha, α — of an object is its reflectivity, i.e., the fraction of incident energy reflected by the object. Explain why we should multiply the Sun’s flux F by a factor of $(1 - \alpha)$, to find the energy absorbed by the Earth. What do you think causes the albedo of the Earth?

⁵The steps we follow here, and in Question 3, are described in Chapter 4 (pp. 53–57 of our textbook by Dessler. See also Figures 4.1–4.4.

Object	Characteristic	Size (m)
Sun	Radius, $R_{\odot}$	7.0×10^8
Earth	Distance to Sun, $d_{\odot \rightarrow \oplus}$	1.5×10^{11}
Earth	Radius, $R_{\oplus}$	6.4×10^6

Table 2: Astronomical distances and sizes

3. **Calculating the temperature of the Earth (Part I)** We found in the previous question that the “solar constant” of the Sun (the solar power received at the position of the Earth in space) is $S = 1360 \text{ W/m}^2$, but that the averaging over the Earth's surface reduces this by a factor of 4, and then the albedo reduces it by a factor of $1 - \alpha$. Therefore the average solar flux at Earth's surface is:

$$F_{\text{Sun on Earth surf}} = \frac{S(1 - \alpha)}{4} \quad S = 1360 \text{ J/s/m}^2$$

We will use this to determine the Earth's temperature by assuming a situation of equilibrium.

- Draw an energy flow diagram for the Earth and the Sun. Ensure that the Earth object is in equilibrium. Is the Sun in equilibrium?
- Write down an equality for the equilibrium that you indicated in your flow diagram, expressed in terms of the power per area (flux):

$$F_{\text{Sun on Earth surf}} = F_{\text{Earth}}$$

The flux of the Sun is given above. what is F_{Earth} (Hint: you have an equation for it in the introductory equations). Using that equation, and an approximate value of $\alpha = 0.3$, solve for the Earth's temperature.

- Is this answer correct? Why or why not? What else might we need to account for?

Greenhouse Effect (Part 1): Heating by Radiation

1. Calculating the temperature of the Earth (Part II)

- (a) We found last week in Discussion (and in Section 4.1 of your textbook) that: (i) The Sun emits a total power of about 4×10^{26} W, where a Watt, W, is a $\frac{\text{J}}{\text{s}}$; (ii) The solar flux at the position of the Earth in orbit is about 1400 W/m^2 (this is called the “Solar constant”, S); (iii) The average solar flux at the surface of the earth, ignoring albedo, is exactly $F_{\text{Sun at surface (ideal avg)}} = S/4$; (iv) The albedo of the earth is about 0.3. Review each of these four statements with your group and then say a few words about each (what it means, how it was calculated, include a picture if it helps).
- (b) Write down an equation for the average flux of the sun at the surface, $F_{\text{Sun at surface, avg}}$, in terms of S and α (i.e., combining (iii) and (iv) from (a)). Explain why it has this form and calculate a numerical value.
- (c) Repeat the calculation from Q3 of Discussion last week (“Part I”) to estimate the Earth’s equilibrium temperature. Recall that we did this by assuming that $F_{\text{in}} = F_{\text{out}}$, where the incoming flux is from the sun (just found in (b)) and the outgoing flux is the Earth’s blackbody radiation, $F_{\text{Earth surf}}$, which depends on its temperature. Discuss with your group the equilibrium temperature value you find and say a few words about it here.
- (d) Let’s construct an “energy flux¹ diagram” associated to the calculation in part (c). Draw a single, wide, rectangular box, representing the surface of the Earth. Draw two flux arrows, one coming into the Earth surface box from above, and one going out of the Earth surface box (also above). Although we had more precise names for these fluxes above, let’s now call them F_{sun} and F_{earth} (label your arrows). Since we have assumed a situation of equilibrium, the Earth’s total energy is constant, so you can indicate that with a horizontal arrow ($\leftrightarrow$) inside the box. Also write a T_{Earth} inside the box to represent this constant energy value, the equilibrium temperature. We know the value for the incoming arrow; write it next to the arrow. Describe how this picture represents the equation in part (c). What must be the value for the outgoing arrow?
- (e) We will now attempt to correct the value found in (c). Let’s start by adding second “object” to our energy flux diagram: the atmosphere.
- Draw two wide rectangles, one above the other, with some separation. Label the bottom “Earth’s Surface” and the top “Atmosphere.”
 - The incoming flux arrow to the Earth surface that you drew in (d) remains. Why does it go into the surface box, not² the atmosphere box?
 - The outgoing flux arrow from the Earth surface also remains, but it’s value might be different in the end. This arrow does not go out into space as before. Why not?
 - The atmosphere should have two flux arrows coming out, each will have the same value, call it F_{atm} . Why does each have the same value?
 - Put “unchanging” arrows ($\leftrightarrow$) inside the boxes. Give a temperature to each box.

¹This is similar to our prior “energy flow diagram” but now the flowing quantity is flux, energy *per area* per time. As you will see in your construction, this will mean that flux is passing from one two-dimensional surface to the next (like the approximately two-dimensional atmosphere above our heads).

²This arrow, from the sun to the Earth could go “behind” the atmosphere box, or around it, but should still come from the top of the page.

- Which flux arrow do we know the number for? Indicate it.
- (f) Let's get some equations from your flux diagram. Use the basic principle of $F_{\text{in}} = F_{\text{out}}$ for a box, to get two equations that relate F_{sun} , F_{earth} , and F_{atm} (two boxes means two equations). Get a third equation by combining the two equations to eliminate one variable.
- (g) Use your equations, and the Stefan-Boltzmann equation, to get a new temperature estimate for the Earth's surface. Is this one correct?

2. **Measuring solar incident flux:** In Lab this week, we measure first the heat capacity of Aluminum and then use that to measure the radiant flux of a desk lamp (or if we're lucky, the Sun) by tracking the temperature increase of an aluminum block over time.

- (a) We find in lab that the aluminum has a specific heat capacity, c_{Al} about 4.5 times smaller than water (0.90 J/g/K vs. 4.2 J/g/K). Write down the defining equation of the specific heat capacity, and say in your own words what it means. Then describe in plain language what it means that the specific heat capacity of water is 4.5 times higher than aluminum (consider, e.g., 10 g of each substance subjected to the same heating). Can you make sense of the word "capacity" in this context?
- (b) Below is a table of data taken for a [class from University of Arizona](#) (where the Sun is a more reliable partner). Let's try to estimate the solar constant³, S , using the data.

- Using graph paper, plot this data (Temperature (C) vs. time (s)) being careful to set the range on each axis so that no space is wasted (look at the minimum and maximum values of each quantity).
- Exam the data and choose a range of times where the temperature seems to be increasing approximately linearly (away from the initial and final "flatness").

Aluminum block dimensions: 7.0 cm x 13.4 cm (93.8 cm²)
 Mass of aluminum block: 324 grams
 Specific heat of aluminum: 0.215 cal/(gram °C)

Time	Elapsed Time	Temperature
9:09:30	0.0 min	27 C (80.6 F)
9:10:00	0.5	27 (80.6)
9:11:00	1.5	27.5 (81.5)
9:12:00	2.5	28.5 (83.3)
9:13:00	3.5	30.0 (86.0)
9:14:00	4.5	32.0 (89.6)
9:15:00	5.5	33.5 (92.3)
9:17:00	7.5	36.5 (97.7)
9:19:00	9.5	39.5 (103.1)
9:21:30	12.0	42.5 (108.5)
9:25:30	16.0	47.0 (116.6)
9:31:30	22.0	52.0 (125.6)
9:38:00	28.5	56.0 (132.8)

- What is the average temperature of the aluminum during that time interval?
- Use a ruler to draw a line of best fit through this portion of the data (you can draw the straight line across your whole graph, but only fit to your chosen data points).
- Find the slope of that line and its units (always use two widely-separated points *on the line* to find a slope, not data points).
- The slope you found is $\frac{\Delta T}{\Delta t}$ for the aluminum block. Now use the defining equation for specific heat capacity and the specific heat of aluminum (0.9 J/g/K) to determine the power gain of the aluminum, $\frac{\Delta E}{\Delta t}$.

³If you point your aluminum block perpendicular to the rays of the sun, and you have a clear sky, then you should be able to estimate S , not the average flux including albedo and the average over the whole Earth's surface!

- Draw an energy flux diagram for the aluminum block. You should have two⁴ flux arrows. In this case, the block is *not* in equilibrium, so you should have an arrow inside the box showing the direction of energy change.
- This diagram implies a non-equilibrium equation of the form:

$$\frac{\Delta E}{\Delta t} = \frac{\Delta E_{\text{in}}}{\Delta t} - \frac{\Delta E_{\text{out}}}{\Delta t}$$

$$\frac{\Delta E}{\Delta t} = F_{\text{block}} A_{\text{block}} - F_{\text{sun}} A_{\text{block}}$$

We calculated the left-hand side number from our graph. The block area, A_{block} , can be calculated from the data in the table (convert to meters). We know the value of the flux of the block outward from its average temperature and the Stefan-Boltzmann law. The flux from the sun is our unknown variable; solve for it!

- Freezing water in the desert night (BONUS in F2025):** When it is nighttime, we don't get the Sun's incoming flux. Luckily, all of the objects around us have retained energy throughout the day, and radiate their own infrared light (heat) to keep us relatively warm until sunrise. If, however, you can restrict an object's line of sight to only the sky — and insulate it thermally to minimize conduction from the ground and air — then you will find that it quickly radiates away its heat into the sky!
 - You made a number of temperature measurements of the sky using your infrared thermometer. What was the average value you (and your classmates) found? Recall that we have the data results in a google survey from the first weeks of class.
 - Using the same equation as the previous question, estimate the rate of energy change, $\frac{\Delta E}{\Delta t}$, of a puddle of water. How long would it take the water to decrease from 20°C to the freezing point?

⁴You could have flux arrows out of the block into the insulating styrofoam, but those will be approximately cancelled by flux arrows into the aluminum from the styrofoam, so we need only include the arrows in/out of the sky-facing surface.

Greenhouse Effect: Matter and Light Interaction (two days)

Here we'll focus on a single topic: how light interacts with molecules to produce the greenhouse effect. This will require us to first review (i) basic physics concepts of electric charges, forces, and fields; and (ii) the basic chemistry concepts (ultimately arising from *quantum* physics) of atoms, bonding, and molecule geometry. We will then put the two together under the basic rule:

Infrared photons can be captured by a molecule if that molecule's net electric dipole can be changed by the photon's electric field. These changes to the dipole will occur concurrently with (because of) vibrations or rotations of the molecules, which is where the energy of the photon goes when captured.

Let's jump right to the point with some example applications of the rule for atmospheric gases:

- **Nitrogen** (N_2 ; $\text{N}\equiv\text{N}$; 78% of the “dry” atmosphere) and **Oxygen** (O_2 , $\text{O}=\text{O}$, 21% of the atmosphere) are each perfectly symmetric linear diatomic molecules. That symmetry is maintained under vibration (stretching and contracting). Therefore they have always have zero net electric dipole. Therefore their dipole moment is unchanging regardless of a photon's electric field. Therefore they do not capture (infrared) photons. This means that 99% of the gases in the (dry) atmosphere are *not* greenhouse gases!
- **Carbon Dioxide** (CO_2 ; $\text{O}=\text{C}=\text{O}$; $425 \text{ ppm} = 425/10^6 = 0.000425 = 0.0425\%$ of the atmosphere) is a symmetric linear molecule. One can imagine two internal electric dipoles pointing along the bonds from C to each O, but these vectors point in opposite directions and sum to a net dipole of zero. If the molecule stretches symmetrically (keeping the two bond lengths equal) then the net dipole moment remains zero (two longer, but still opposite, vectors sum to zero). However, if one C=O bond is stretched more than the other, then there *will* be a net dipole. So infrared photons can be captured to cause that type of (asymmetric) vibration. Additionally: if the molecule *bends*, the net dipole will be nonzero; therefore it can absorb infrared photons to induce bending oscillations. Carbon dioxide is a greenhouse gas.
- **Methane** (CH_4 ; $\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{H} \\ | \\ \text{H} \end{array}$; about $0.0002\% = 0.000002 = 2 \text{ ppm}$ of the atmosphere) has a perfectly tetrahedral structure (with bond angles of $\approx 109.5^\circ$). Like CO_2 : (i) it has a zero net dipole moment (because the three imagined dipole vectors along the bonds sum to zero); (ii) symmetric vibration/stretching (maintaining equal bond lengths) will maintain a zero net dipole moment; but (iii) it can have a *non-zero* dipole moment if there is *asymmetric* stretching of the bonds (any of the four bond lengths different than others). Therefore it can also capture infrared photons to cause (asymmetric) vibrations. Methane is a greenhouse gas.
- **Water vapor** (H_2O ; $\text{H}-\text{O}-\text{H}$; zero to 4% of the total atmospheric gases depending on conditions, 0.4% on average) also has a tetrahedral electron domain structure (formed by a central oxygen with two hydrogen atoms and two electron pairs), but it is not a perfectly symmetric tetrahedron like methane. As a molecule, the two hydrogen “points” of the tetrahedron together with the central O form a bent molecule: $\text{H}-\overset{\curvearrowright}{\text{O}}-\text{H}$. It has a net dipole at rest, whose magnitude can change under vibrations and rotations. Therefore it can capture infrared photons to induce those vibrations and rotations. Water is a greenhouse gas.

Our goal is that, by the end of the exercises below, the above rule and the details of the examples will make sense (they certainly might not make sense right now!).

1. Electric charge, Electric Field, Dipoles, Light-Dipole Interaction

- (a) **Electric charge** Observation of macroscopic objects demonstrates that there are two types of charges (we now call them positive and negative, and know that they arise, at the atomic level, from protons and electrons, respectively). When two different objects are rubbed together, one is more likely to give up electrons than the other, and therefore becomes positively charged, while the other becomes negatively charged. It is seen that these two objects are then attracted to each other. Play with some of the objects coming around in the demo box to see this in action. Then see if you can understand why one must conclude that there are two types of charges (e.g., can you figure out how to make two objects repel each other?).
- (b) **Electric Force** We summarize the interaction between two charged objects with the *electric force law* (or, Coulomb's Law). It states that like charges repel, while unlike charges attract (the strength of the force drops off with the charge separation). In each of the two example pictures below (separated by the horizontal line), draw a force vector on each of the two charges in that picture. Label the charges q_1 and q_2 , and label the forces $\vec{F}_{q_1 \text{ on } q_2}$ and $\vec{F}_{q_2 \text{ on } q_1}$.



- (c) **Electric Field** Physicists decided that they did not like “instantaneous action at a distance” forces, so they introduce a new concept called the *Electric Field*. Each charge produces its own electric field everywhere in space (with changes to the field, induced by motion of the charges, propagating outward at the speed of light). Then another charge can sit in the field of the first charge and experience a force due to the field (see next question).

For a positive charge, the electric field points outward in all directions, and is drawn as equally spaced lines emanating outward with arrowheads in the middle of those lines (not at the ends like an arrow/vector). Negative charges have fields that look the same, but the arrowheads point inward. Once you draw that field, you could go to any point in space, draw a dot, and draw *the electric field vector* at that point, $\vec{E}$, which is a vector/arrow that is parallel to the field lines at that point. Draw the electric field lines in each of the two pictures shown below. Make few points in space, and draw the electric field vectors at those points (these will be larger/stronger closer to the source charge).

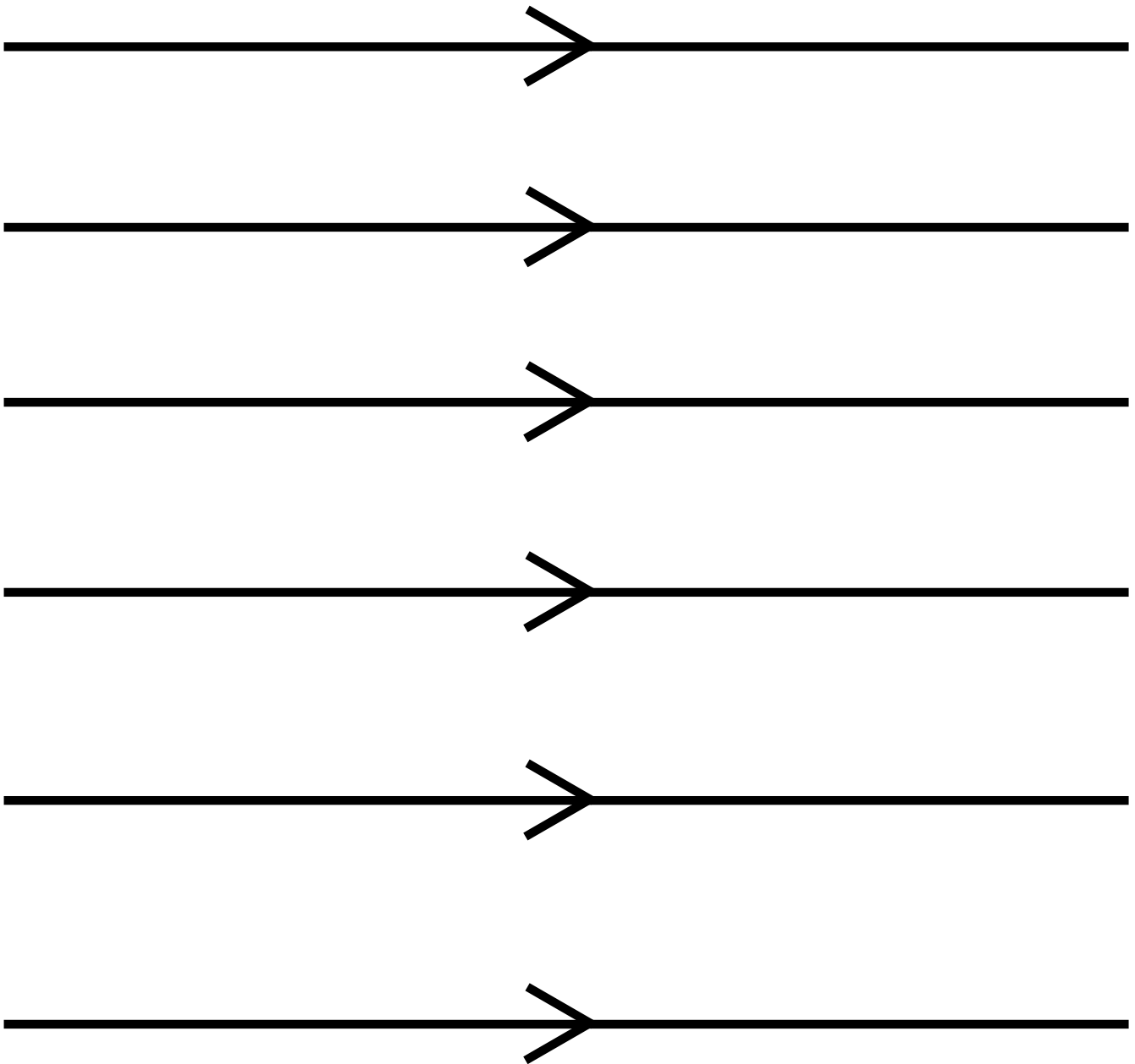


- (d) **Electric Field Force** The electric force law is given as $\vec{F}_{E \text{ on } q} = q\vec{E}$. This means that the force vector on a charge will *point the same direction as the field when the charge is positive*, but *the opposite direction when the charge is negative*.

Now consider the again the two pictures of charge pairs (reproduced below). In each picture, draw: (i) the field of the left charge (q_1) everywhere in the picture; (ii) the electric field vector, $\vec{E}_1$, at the position of the second charge; (iii) The force vector on the second charge, $\vec{F}_{E_1 \text{ on } q_2}$.

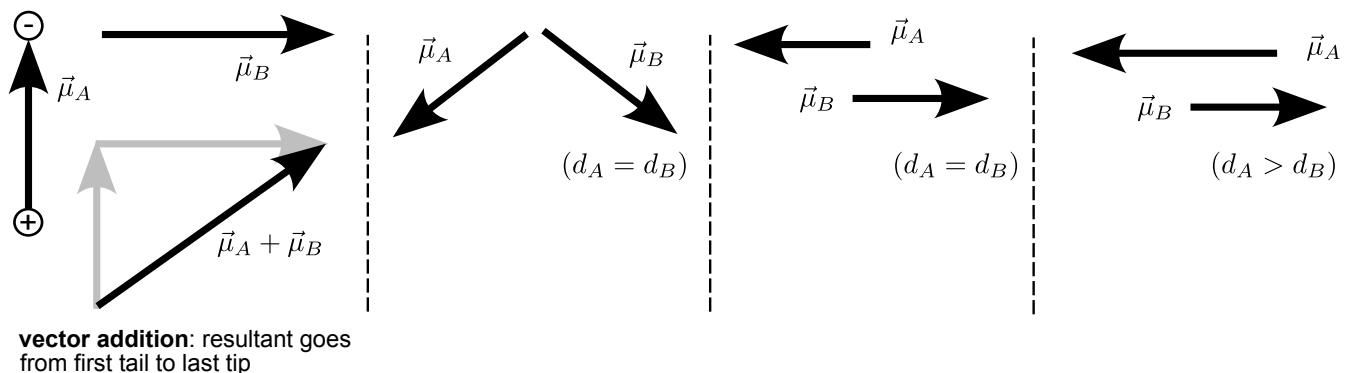


- (e) Now that we have the electric field, we can draw it and consider its effect on charges without considering the source charges that produced it. Consider the *uniform* electric field below (same magnitude of $\vec{E}$ everywhere). Draw a few little positive and negative charges in the field. Indicate the $\vec{E}$ vector at their location. And then draw the $\vec{F}_E$ on q force vector on each charge.



- (f) **Electric Dipole** An electric dipole can be thought of as: two opposite charges, q and $-q$, separated by a rigid spring (with relaxed length, d). Fill the space below with a uniform electric field (like the previous page) and then draw multiple dipoles oriented in various orientations with respect to the field. Indicate the force on each charge in the dipole. Describe the motion induced by the forces, if any¹

We represent an electric dipole schematically as a single vector, $\vec{\mu}$, that points from the positive to the negative charge, whose magnitude (strength) is $|\vec{\mu}| = |q|d$. The sum of multiple dipoles creates a *net electric dipole*, and is found by **vector addition**. Consider the “tail to tip” addition method shown at left and then try it for the other cases.



¹Remember, forces cause *accelerations*, meaning *changes in velocity*. So if a dipole is placed in a field at rest, forces could cause it to *start moving* — from zero velocity to some velocity.

- (g) **Electromagnetic Waves** To represent an electromagnetic (EM) wave traveling across the page from left to right: (i) draw an x - y axis, with origin at left (long x axis, short y); (ii) draw a sinusoidal wave along the x axis [$y(x) = A \sin(x)$, so half in positive y , half in negative] with about three wavelengths spread across the page (show the length, λ), (iii) draw a velocity vector anywhere near the wave, labeled $\vec{v}$, pointing to the right.

Light is an *electromagnetic wave* and the y direction of the wave you drew is meant to represent the *electric field* magnitude and direction (we will ignore the magnetic field part of the EM wave).

Add the electric field to the picture above: (iv) In between the x axis and the sine curve, draw vertical electric field vectors with their tails on the axis. They point in the positive y direction when the curve is positive, and in the negative direction when the curve is negative. The vectors have zero length twice per wavelength. This represents a *plane wave* of light, so the electric field vector at one position on the x axis actually represents the electric field at all locations on an (infinite) 2D plane perpendicular to the axis there.

In the space below: (v) Using your plane wave above, now draw an electric field that fills the space. Use many $\vec{E}$ vectors (not “field lines” like prior questions). Along any vertical line you should have multiple $\vec{E}$ vectors of the same length. (vi) Draw some electric dipoles, sitting in this field and indicate the forces on their pairs of charges.

(vii) Look up the size of a few molecules listed on the first page. We found earlier that the wavelength of infrared light is about one to 10 micrometers ($1 \mu\text{m} = 10^{-6} \text{ m}$). How do these two sizes compare? As an EM wave passes by a molecule, is there a lot of variation of the $\vec{E}$ field strength across the molecule, or a little?

2. Chemistry: Atoms, Molecules, Geometry, Dipoles

Undergraduate physics majors take a course in Quantum Mechanics that culminates in constructing the discrete energy levels of a hydrogen atom, i.e., the possible excitations of a single electron bound to a single proton. These energy levels are labeled by nested integers²

$$\begin{aligned}n &= 1, 2, 3, \dots \\ \ell &= 0, 1, 2, \dots, n-1 \\ m &= -\ell, -\ell+1, \dots, 0, \dots, \ell-1, \ell \\ m_s &= -1, +1\end{aligned}$$

where each energy level has a unique set of (n, ℓ, m, m_s) , i.e., you choose one number from each row above. For example, when hydrogen's electron is in the "ground state" (lowest energy level), it has $n = 1$. This means that $\ell = 0$ (since ℓ only goes up to $n - 1$), which means that $m = 0$ (since m only goes up to $\pm\ell$), and then m_s can be either ± 1 . So the ground state of hydrogen is $(n, \ell, m, m_s) = (1, 0, 0, -1)$.

It turns out that other atoms can be approximately constructed by "filling" the energy levels of hydrogen — called the *orbitals* of hydrogen — with additional electrons (and increasing the protons in the nucleus by the same number³). So, hydrogen with one electron fills the $(1, 0, 0, -1)$ level, but helium has two electrons and fills two levels $(1, 0, 0, -1)$ and $(1, 0, 0, +1)$. These are the only two levels with $n = 1$, so the three-electron atom, lithium, fills the two levels of helium plus the level $(n, \ell, m, m_s) = (2, 0, 0, -1)$.

Chemistry uses this orbital model to build molecules⁴ from atoms. The first two rows of the periodic table (see next page) correspond to complete the $n = 1$ and $n = 2$ levels, respectively, and the third row corresponds to $n = 3$ with $\ell = 0, 1$ ($\ell = 2$ appears on the fourth row, and we will ignore for now). Each of these rows is considered a "shell" which is "closed" by reaching the end of a row. The electrons in the "outermost shell" are called *valence* electrons. The first row of the periodic table has up to two valence electrons, while the second and third rows have up to eight valence electrons (corresponding to the eight possible ℓ and m values). Molecules are formed following these rules (see also the figures on the next page):

- Electrons are placed on atoms in a *Lewis dot* convention, with four pairs in the four cardinal directions (N, S, E, W). For H and He (the first row, $n = 1$, case), one pair.
- As the valence electrons are filled up (moving across a row in the periodic table), each pair receives one electron first, before pairs are filled with two electrons.
- If two or more atoms form a molecule, each atom will "fill" its valence shell (obtain eight electrons),

²The physical meaning of these labels correspond, approximately, to breaking the electron's three-dimensional motion around the proton into radial movement (n), and angular movement (ℓ, m). The fourth label, m_s corresponds to the "spin" of the electron.

³Neutrons, the other particle in the nucleus of an atom, come along as well, with their number dictating the atomic mass. An atom is defined by the number of protons (or electrons); changing the number of neutrons in an atom just changes the "isotope." Atoms naturally occur with different numbers of neutrons. The atomic mass listed on the periodic table is an average over their frequency, which is dictated by the rules of nuclear physics and stability.

⁴Here, we will consider only *covalent bonding*, where electrons are shared between neutral atoms. Another type of bond, an *ionic bond*, is formed by the attractive electric force of positive and negative *ionized atoms* (atoms that have lost or gained one or more electrons, e.g., salt: Na^+ plus Cl^- equals NaCl).

- (a) Form molecules for: N_2 , O_2 , CH_4 , CO_2 , and H_2O . Use the rules described above and take these steps:
- Draw the Lewis dot structure for each atom that will join to form the molecule first.
 - Arrange the atoms next to each other so that unpaired electrons can be shared by neighboring atoms.
 - Replace any shared pair of electrons with a solid line, leaving each atom's own pairs as two dots.
- (b) Use the potatoes provided to physically build these molecules following your construction in the previous question. Use big yellow potatoes for O, N, C, and little yellow potatoes for H. Use little red potatoes for bonds and free electron pairs. Hold everything together with sticks. Try to arrange these such that the red potatoes are pushed furthest apart (this is the electric force repulsion of electrons).
- (c) Given your physical structures, determine why the molecular structure of the greenhouse gas molecules are as described on the first page.

	1A	2A	3A	4A	5A	6A	7A	8A		I	II		III	IV	V	VI	VII	0
n	H 1							He 2		H •								He ••
1	•							•		Li •	Be ••		B •	C ••	N •••	O ••••	F •••••	Ne ••••••
2	Li •	Be ••	B •	C ••	N •••	O ••••	F •••••	Ne ••••••		Na •	Mg ••		Al •	Si ••	P •••	S ••••	Cl •••••	Ar ••••••
3	Na •	Mg ••	Al •	Si ••	P •••	S ••••	Cl •••••	Ar ••••••		K •	Ca ••		Ga •	Ge ••	As •••	Se ••••	Br •••••	Kr ••••••
										Rb •	Sr ••		In •	Sn ••	Sb •••	Te ••••	I •••••	Xe ••••••
										Cs •	Ba ••		Tl •	Pb ••	Bi •••	Po ••••	At •••••	Rn ••••••

• Metal
• Metalloid
• Nonmetal

Atom	$N_{e^-} = N_p$	Energy levels filled	“Electron Configuration”
H	1	(1, 0, 0, -1)	$1s^1$
He	2	(1, 0, 0, -1), (1, 0, 0, +1)	$1s^2$
Li	3	He + (2, 0, 0, -1)	$1s^2 2s^1$
Be	4	Li + (2, 0, 0, +1)	$1s^2 2s^2$
B	5	Be + (2, 1, -1, -1)	$1s^2 2s^2 2p^1$
C	6	B + (2, 1, 0, -1)	$1s^2 2s^2 2p^2$
N	7	C + (2, 1, +1, -1)	$1s^2 2s^2 2p^3$
O	8	N + (2, 1, -1, +1)	$1s^2 2s^2 2p^4$
F	9	O + (2, 1, 0, +1)	$1s^2 2s^2 2p^5$
Ne	10	F + (2, 1, +1, +1)	$1s^2 2s^2 2p^6$
Na	11	Ne + (3, 0, 0, -1)	$1s^2 2s^2 2p^6 3s^1$
Mg	12	Na + (3, 0, 0, +1)	$1s^2 2s^2 2p^6 3s^2$
Al	13	Mg + (3, 1, -1, -1)	$1s^2 2s^2 2p^1 3s^2 3p^1$
Si	14	Al + (3, 1, 0, -1)	$1s^2 2s^2 2p^2 3s^2 3p^2$
P	15	Si + (3, 1, +1, -1)	$1s^2 2s^2 2p^3 3s^2 3p^3$
S	16	P + (3, 1, -1, +1)	$1s^2 2s^2 2p^4 3s^2 3p^4$
Cl	17	S + (3, 1, 0, +1)	$1s^2 2s^2 2p^5 3s^2 3p^5$
Ar	18	Cl + (3, 1, +1, +1)	$1s^2 2s^2 2p^6 3s^2 3p^6$

3. Interaction of Light with Molecules: Vibrational Modes, Dipoles, Rotation

- (a) Assign a dipole vector to each red potato on your physical models (or if less daring, just try to draw on your written models).
- (b) Imagine possible rotations and vibrations of each molecule and try to understand:
 - i. Why are water, methane, and carbon dioxide greenhouse gases?
 - ii. Why are oxygen and nitrogen not greenhouse gases?

Greenhouse Effect: Infrared Absorption Spectrum

In our work over the last week, we investigated how the structure of molecules determines their potential to act as greenhouse gases (absorbing the infrared radiation of the Earth's surface). We found that:

a molecule can absorb infrared photons if there is some oscillation of the molecule (vibration, bending, rotation) that causes its net electric dipole moment to change

If such an oscillation exists, then passing infrared photons will be absorbed by the molecule, and the energy they deliver will “excite” the oscillation. We saw that a “changing dipole moment” is possible if one or both of the following is true:

- (i) The molecule has a net electric dipole moment (when relaxed), which can be altered by the electric field of a photon of electromagnetic radiation.
- (ii) The molecule has a net zero electric dipole moment when relaxed, but some non-relaxed configurations have a non-zero dipole moment; these oscillations — accompanied by a changing dipole moment — can be induced by infrared photons.

Both of these can occur together (e.g., in the water molecule), or just one (e.g., carbon dioxide, CO₂, and methane, CH₄, are both symmetric molecules and so they have zero net dipole moment; but when their bonds are stretched asymmetrically, they do have a dipole moment).

Today we will investigate how the absorption of infrared photons in Earth's atmosphere causes a loss of energy reaching space, which is observed as so-called **absorption lines** in the blackbody spectrum. The key ideas of absorption are these:

- The energy of light is carried in photons, which are packets of light of a single color (a single wavelength or frequency). The amount of energy in a single photon is given by the equation:

$$E = hf = h \frac{c}{\lambda}$$

where $h = 6.6 \times 10^{-34}$ m²kg/s is called Planck's constant, f is frequency, and λ is wavelength. This means that a single 6 μm wavelength photon has just 33×10^{-19} Joules of energy (33 “zeptojoules”!). To simplify our calculations, we will use this easy conversion between frequency in terahertz (1THz = 10¹² Hz) and wavelength in micrometers:

$$f [\text{THz}] = \frac{300}{\lambda [\mu\text{m}]}$$

- A type of “oscillation” is called a “mode” or “vibratory state” of the molecule. Its excitation has a certain fundamental energy associated to it, and, if accompanied by a changing electric dipole, it will be excited by photons of a specific frequency/wavelength (see Figure 1 and Question 2; in Figure 1 these wavelengths are labeled by “ ν ” instead of “ λ ” for some reason).
- Combinations of multiple oscillations can also occur. The sum of their excitation energies can then be associated to a single photon frequency that can excite the combination. A special case of this is a “double excitation” (or triple, ...) of a single oscillation type; so, if f is an excitation frequency, then $2f$ is also an excitation frequency. (See Question 3).

1. **Electric dipoles and absorption** For each of the three greenhouse gas molecules shown in Figure 1, complete the following steps:

- Draw the structure for yourself, reminding yourself why it looks as it does, using our “chemistry” work in the previous discussion¹.
- On one picture of the molecule, draw an electric dipole vector along each bond. If there are identical types of bonds (e.g., the two carbon to oxygen bonds in CO₂) then they should have identical length vectors. Using the rules of vector addition that we learned on the previous discussion, determine if the resting molecule has a net dipole moment or a zero dipole moment.
- Now consider the three “vibratory states” (also called *modes*) shown in Figure 1 (vibration along bonds, or bending of the molecule) . Explain each of these oscillations with a *sequence of pictures*, showing how the structure changes over time, periodically.
- Under the oscillations shown in the previous step, will the net dipole moment change? Check your answers² using the information given in Figure 1.

2. **Drawing absorption spectra on blackbody radiation curves**

- (a) Recall from Week Four’s discussion how we drew blackbody curves on log-log paper. Sketch the Earth’s blackbody spectrum on the log-log paper provided, assuming an average temperature of $T_{\text{Earth}} = 288 \text{ K} = 15^\circ\text{C} = 60^\circ\text{F}$. Remember this involves finding: the wavelength of the peak, λ_{peak} ; the maximum value of the spectral flux; and wavelength positions ($\lambda_{\text{peak}}/2$ and $3\lambda_{\text{peak}}$) of the “ $\frac{\text{peak}}{10}$ ” values of spectral flux . Calculate all values first before setting the scale on your axes, so that you can fill the whole page. The horizontal axis should be the wavelength in units of micrometers, $1 \mu\text{m} = 10^{-6} \text{ m}$.
- (b) For each of the three greenhouse gases in Figure 1, find the wavelength of infrared radiation that excites each mode (these are given in Figure 1). Draw a dot on your blackbody curve for each of these (you should have 8 dots) and put a nearby label with the name of the gas and type of vibration.
- (c) Consider Figures 2 and 3. Find the same eight infrared absorption wavelengths on one or both of these plots and mark them with a vertical line across all sub-graphs. Do these lines make sense based on the information being presented in the figure? Explain.
- (d) Figure 2 shows that most of these lines absorb almost 100% of outgoing infrared light at their peak wavelength. Explain how this is seen in the graph(s). Now, choose a few of the wavelengths (two or three). For each, identify four wavelength values from Figure 2: (i) the values above and below the center line wavelength at which the absorption rises above 0%, and (ii) the two values at which absorption reaches 90% (where the emission is an 10%).
- (e) Using the four absorption wavelengths values you found in (d), draw approximate absorption “valleys” on your own blackbody curve. Draw two dots on your blackbody curve for the (i) values, and then go down by a factor of 10 (one major tick mark on the logarithmic axis), and plot the (ii) values.

¹We didn’t consider Ozone (O₃) before. To figure out its Lewis dot structure, first consider the *total number* of valence electrons it has across all three atoms. Then assign these electrons to the three-atom structure in such a way that each atom gets eight electrons (either in lone pairs or shared). You should find a single bond and a double bond.

²The answer is “yes” if there is a wavelength assigned to the oscillation in Figure 1 (it is yes for eight of the nine).

3. **Additional absorption lines** You might have noticed that there are many absorption lines (places where infrared light is strongly absorbed) shown in Figure 2 that were not accounted for in the eight fundamental absorption lines for water, carbon dioxide and ozone. Many of the lines can be determined from *multiple-oscillation excitations* whose total energy is provided by a single photon corresponding to that total energy value. This can be double- or triple-excitations (etc) of a single oscillation mode, where a single photon puts twice or three times the amount of energy into the oscillation. Or, it can be a true multiple-oscillation mode (e.g., a combination of bending and stretching) that is excited by a single photon with energy equal to the sum of the constituent oscillation energies.

Let's start with water (we'll come back to CO₂ and ozone if we have time) and complete the following steps.

- (a) Calculate the frequency (in terahertz, THz) for each vibratory state from its wavelength in micrometers³. Call these f_1 , f_2 , and f_3 , corresponding to the λ values in Figure 1.
- (b) Make a table with columns: [n_1, n_2, n_3 , sum-of-freq⁴, λ].
 - i. You can fill in the first row of your table with $n_1 = 1$, $n_2 = 0$, and $n_3 = 0$ to represent the first vibratory state; then the "sum-of-freq" will just be the f_1 value and the wavelength will be that of the first vibratory state. Do the same in the next two rows, using the information for the second and third vibratory states. These first three rows correspond to the modes we already found in the absorption lines of Figures 2 and 3.
 - ii. Now, in the fourth row, consider the multiple-oscillation case of combined "symmetric stretching" plus "bending." You will put $(n_1, n_2, n_3) = (1, 1, 0)$ for the n values, and the "sum-of-freq" column will show " $f_1 + f_2 = \dots$ ". Finally convert that total frequency value to a wavelength in micrometers.
 - iii. Complete some more rows either by combinations of different oscillations, or multiple excitations of the same oscillations, e.g., $(n_1, n_2, n_3) = (2, 0, 0)$.
- (c) For each row in your table, identify the locations of the absorption lines on the water vapor graph within Figure 2. Did you find all of them? Are there any that you predicted that you did not find?
- (d) Why are there two carbon dioxide absorption lines between 2 μm and 3 μm ? What are their exact wavelength values?

³See the simple conversion equation on the first page of this discussion. Unhelpfully, the wavelengths in Figure 1 are labeled with the Greek letter "nu" instead of "lambda": ν is given in micrometers, so just think of it as λ .

⁴The sum of the frequencies will be proportional to the energy of the combined oscillation because of $E = hf$, as described on the first page. So the actual procedure should be (i) Calculate the frequency of the vibratory state from its wavelength. (ii) Calculate the energy of the vibratory state from its frequency, using $E = hf$. (iii) Sum the energies of the combined vibratory states. (iv) Calculating the frequency associated to that combined energy using $E_{\text{tot}} = hf_{\text{tot}}$. (v) Calculate the wavelength associated to f_{tot} . But because of that proportionality between energy and frequency we can skip steps (ii)-(iv)!

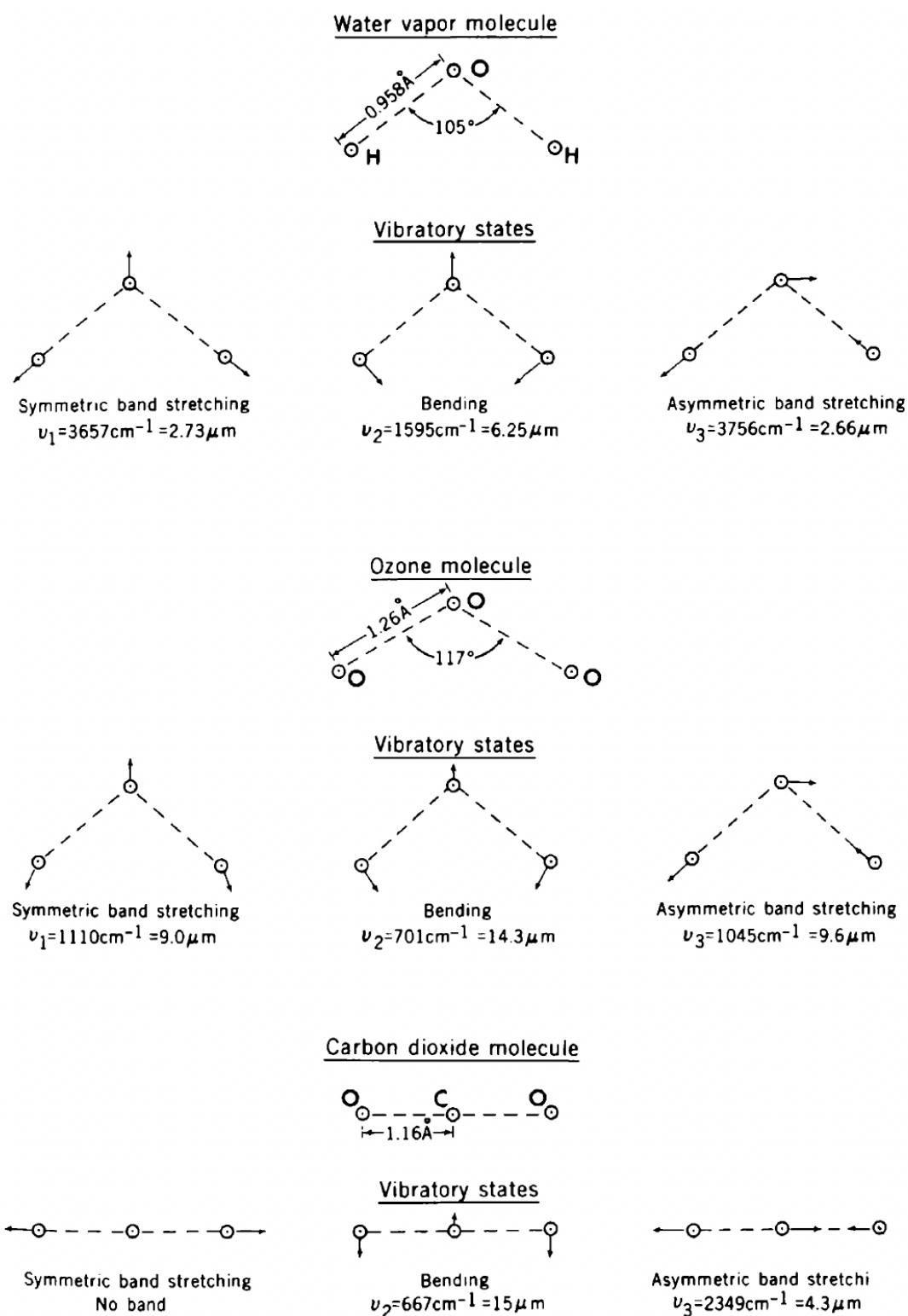


FIGURE 6.5. The structures of the water vapor, ozone, and carbon dioxide molecules and their vibratory states (from Houghton, 1985).

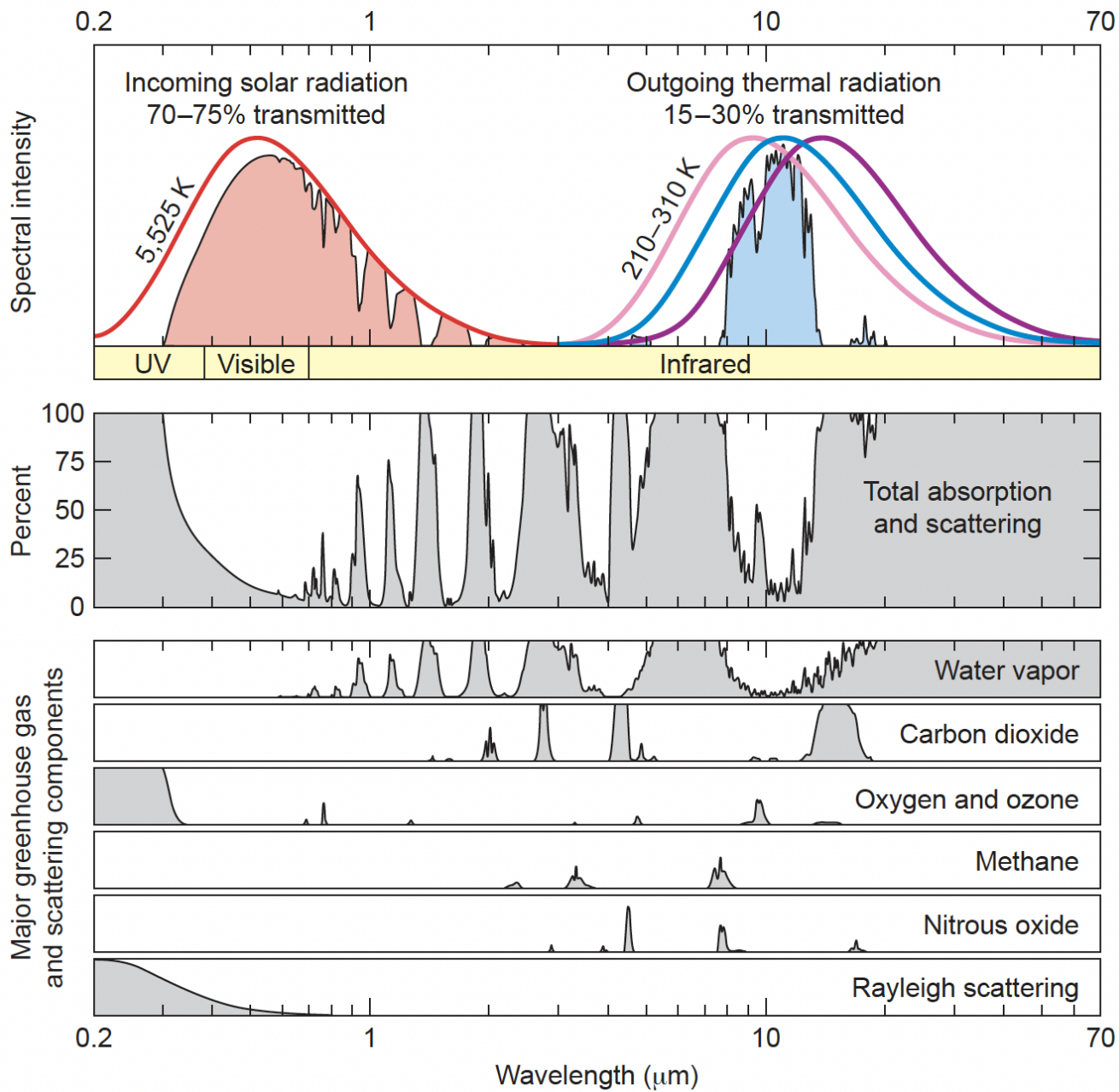


Figure 2: Figure by Rohde, from Mathez textbook

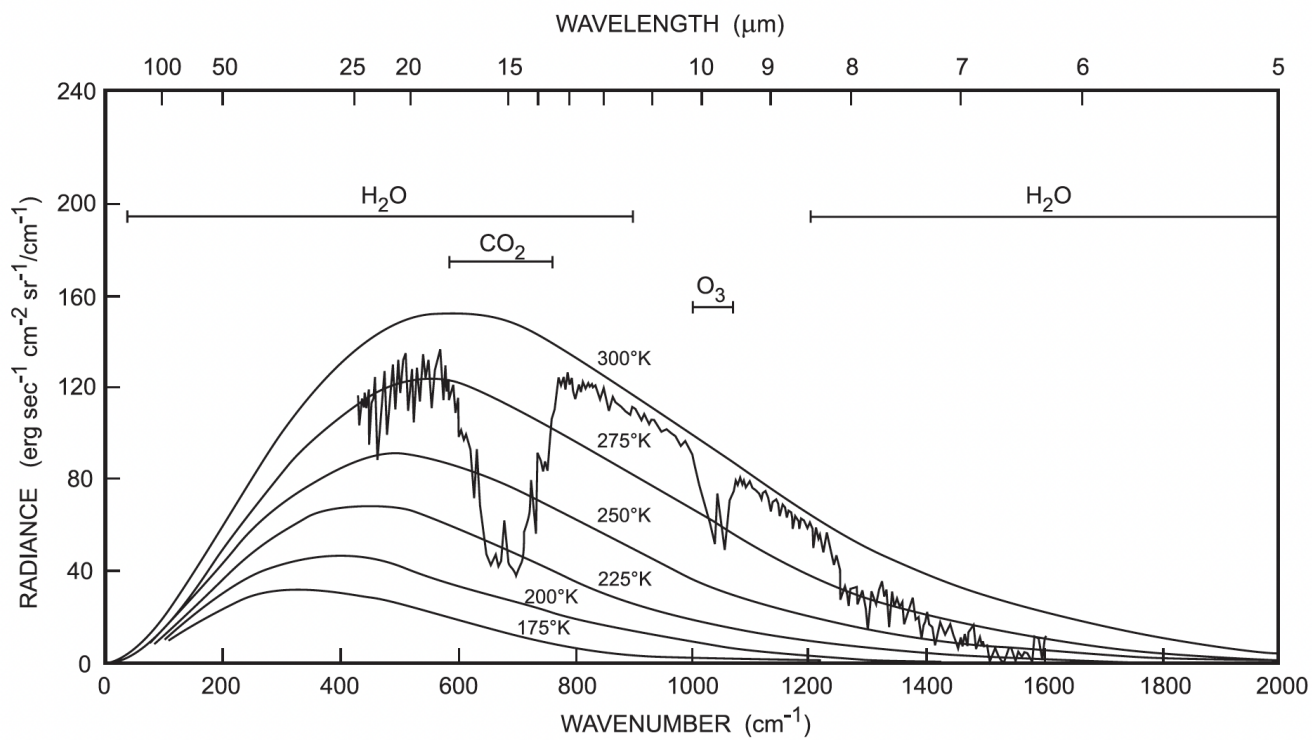


Figure 8.5 Spectrum of outgoing LW radiation over (215 W, 15 N) observed by Nimbus-4 IRIS, as a function of wavenumber λ^{-1} . Blackbody spectra for different temperatures and individual absorbing species indicated. Adapted from Liou (1980).

Figure 3: From Salby textbook

Evaporation, dew point, fog, frost

In lab this week we examine the **psychrometric chart** — relating relative humidity to absolute humidity — and the **phase diagram** for water — depicting the coexistence curve between liquid water and water vapor near 0°C. Both are reproduced in this packet on a later page, along with the larger phase diagram for water that we saw in Lab 07.

The smaller phase diagram is useful for understanding the evaporation of water exposed to air, because in that case the relevant pressure is not the atmospheric pressure, p_{atm} , but instead the **partial pressure**, e , defined as

$$e = f \times p_{\text{atm}} = \left(\frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{\text{dry air}}} \right) \times p_{\text{atm}} ,$$

where you can see that partial pressure is found by multiplying the atmospheric pressure by f , the **mole fraction of water in air**. In lab you re-write this mole fraction in terms of the **absolute humidity**, AH. That new expression is:

$$\begin{aligned} e &= \left(\frac{\frac{m_{\text{H}_2\text{O}}}{m_{\text{dry air}}} \cdot \frac{M_{\text{dry air}}}{M_{\text{H}_2\text{O}}}}{\frac{m_{\text{H}_2\text{O}}}{m_{\text{dry air}}} \cdot \frac{M_{\text{dry air}}}{M_{\text{H}_2\text{O}}} + 1} \right) \times p_{\text{atm}} \\ &= \left(\frac{\text{AH} \cdot 1.6}{\text{AH} \cdot 1.6 + 1} \right) \times p_{\text{atm}} \end{aligned}$$

where the mass ratio $\frac{m_{\text{H}_2\text{O}}}{m_{\text{dry air}}}$ is the absolute humidity read from the vertical axis of the psychrometric chart (but expressed as a pure ratio, e.g., 7 g/1 kg = 7 g/1000 g = 0.007), and where the ratio of the molar masses is $M_{\text{dry air}}/M_{\text{H}_2\text{O}} = 28.8/18 = 1.6$. So now you can:

- Look up or measure the “weather conditions”: relative humidity (%) and ambient temperature.
- Use the psychrometric chart to convert these to an absolute humidity value, AH.
- Calculate the associated partial pressure of water in air, using the final equation above.
- Determine the vaporization temperature (liquid to vapor) from the phase diagram.

This vaporization temperature — which is called “the boiling point” in a *pure* water liquid-vapor transition — is called the **dew point** when considering water exposed to air.

1. **The dew point and water evaporation** Let's calculate the dew point temperature for a few different locations with diverse weather conditions. In each case, draw a horizontal line on the phase diagram provided corresponding to your calculated partial pressure (label the line its location/conditions). Draw a dot on the line where it crosses the coexistence curve and a star on the line at the associated ambient temperature value.
 - (a) Let's start with "indoor conditions." Modern heating and air conditioning is designed not only to keep the temperature at "room temperature" (look this up in °C), but also a comfortable 50% relative humidity.
 - (b) Now let's do Allendale/Grand Rapids yesterday in the afternoon. You can google for "NWS 7 day grand rapids" and then navigate to the "3 day History" link (remember to convert temperatures from Fahrenheit to Celsius... google can do this for you). Do you remember, was it comfortable outside (both in terms of temperature and humidity)?
 - (c) On the search box of the NWS page, navigate to a different city where you expect that conditions were less humid than here. Again look up their afternoon data. Alternatively, you might look for a very dry day in a very dry city¹.
 - (d) Repeat the previous, but for a location with very humid conditions.
 - (e) Use the website in the footnote for (c) to find the conditions in Grand Rapids on a typical winter afternoon (e.g., in January). In these conditions (when temperature is below 0°C) you can still find an absolute humidity from the psychrometric chart, but will need to look at the larger phase diagram. You will find that there is no longer a transition from liquid to vapor, nor a dew point, but there is a "frost point." Find that value and discuss its meaning.
2. **Why dew?** Using our understanding of evaporation developed in the previous question and in lab, let's try to answer some qualitative questions about our surroundings. Feel free to look for advice from your instructor or the web, but try to give answers that reference the technical understanding we have from the phase diagram.
 - (a) In lab this week we do an experiment where the temperature of a beaker of water is slowly lowered by the introduction of ice. What did you observe (or would you expect to observe) on the outside of the beaker and when?
 - (b) Look up the temperature of a standard refrigerator. If you remove a jar of olives from the fridge, do you expect to see condensation? Where does the "condensation" come from?
 - (c) Look up the temperature of a standard freezer. If you remove a tub of ice cream from the freezer, do you expect to see condensation? What do you expect to see and why?
 - (d) Why is there dew on your car on some mornings and not on others? Think about this question in terms of the dew point you worked with in question 1 and in terms of condensation. Is the temperature of the air the same as the temperature of the car? How can this occur? Where does the dew come from?

¹For example, you could google for "when is it hottest and driest in Phoenix" and then use a historical weather database like <https://www.wunderground.com/history> to obtain their afternoon temperature and humidity, under "Daily Observations."

- (e) If I have dew on my car windshield I usually find that using the wipers clears it up only for a moment. Why is that the case, and what controls should you use to clear it up completely?
- (f) When does frost appear on your car?
- (g) What does it mean when you “see your breath”? Remember that water vapor is invisible, and clouds are formed when the vapor condenses into tiny liquid droplets (or ice particles).
- (h) On a cool fall morning, I saw mist rising off a pond on my drive in to work. Why is this happening?
- (i) What is fog and what are the conditions for it to occur?
- (j) Examining your responses to Question (1), what does relative humidity seem to indicate? Why is this measure important to us in terms of how we feel on a given day? Why is a high relative humidity uncomfortable?

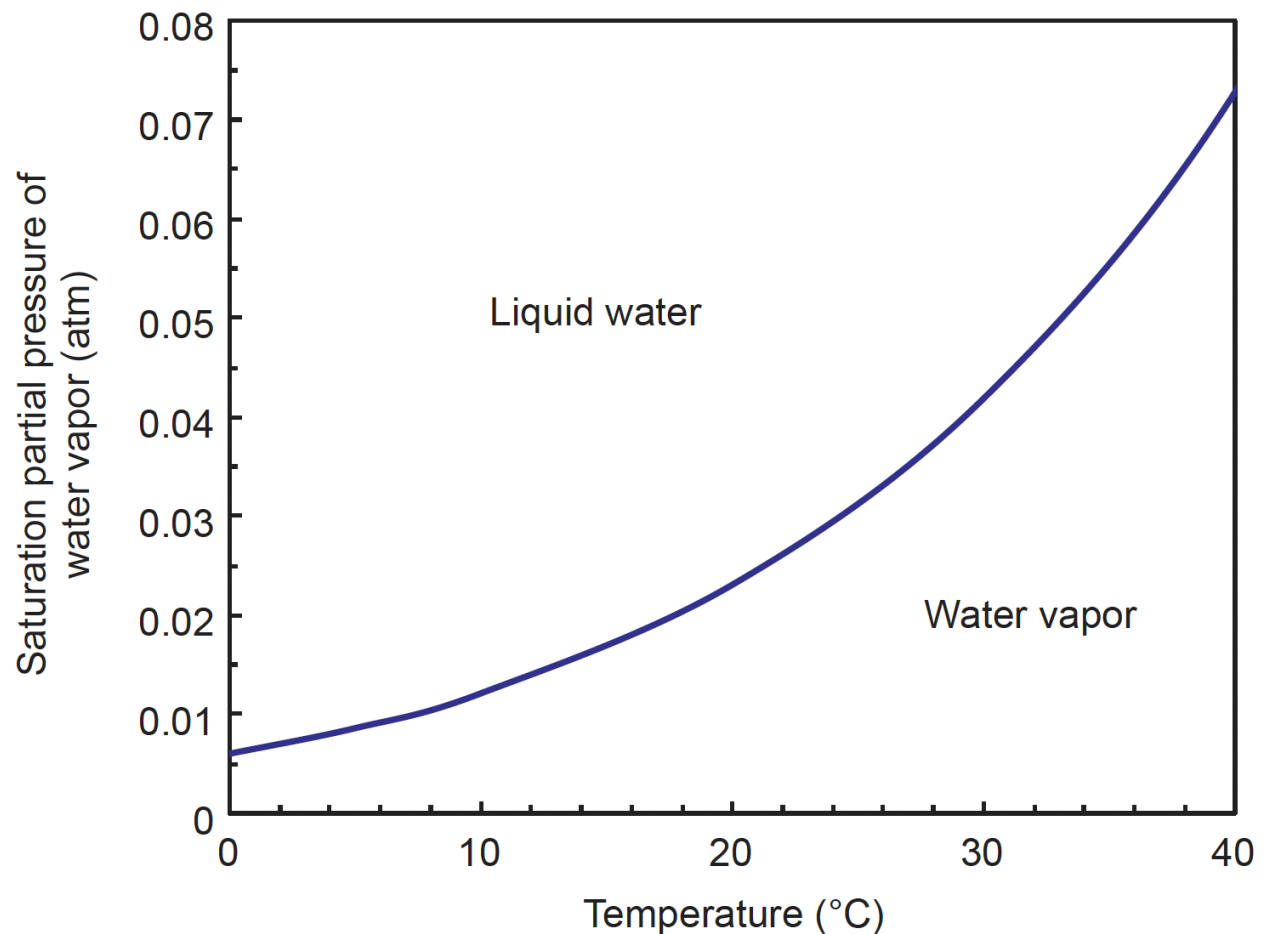
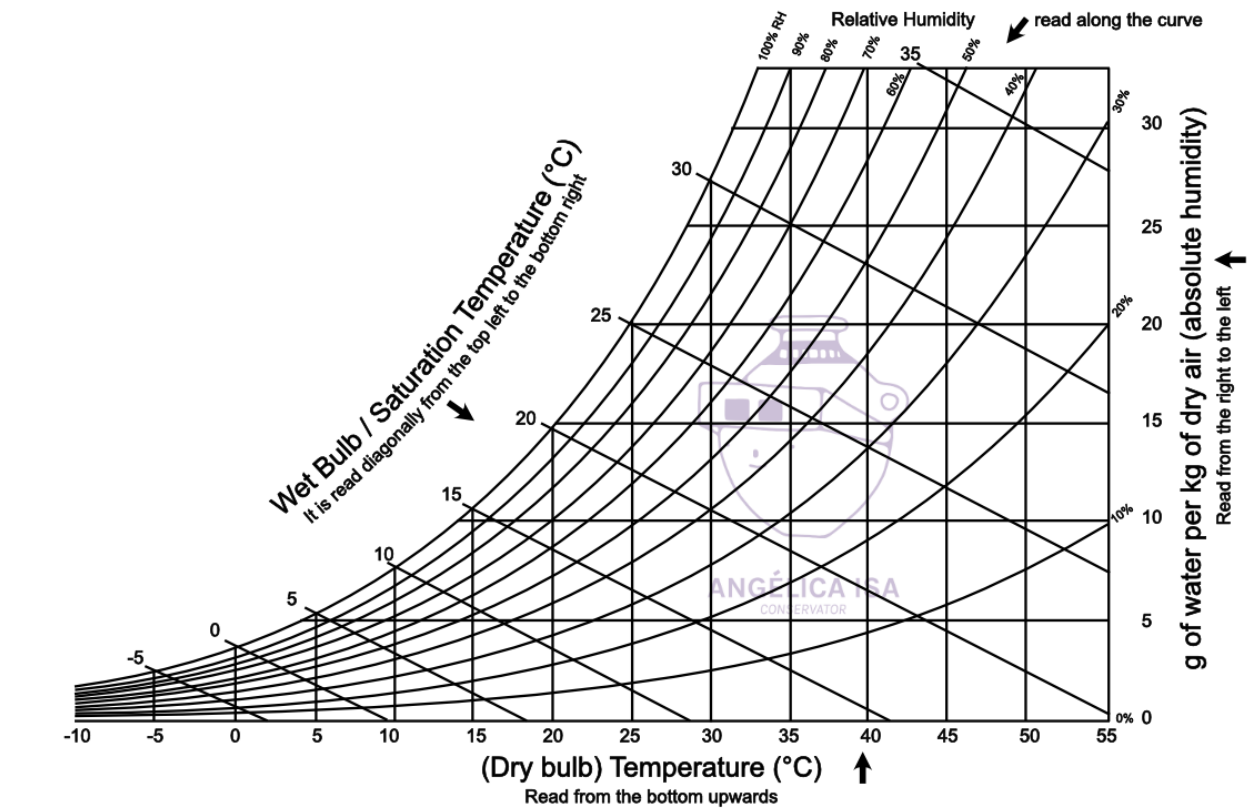


Figure 1: **Top:** Psychrometric chart for water at standard pressure conditions 1 atm = 101 kPa (from angelicaisa.com). **Bottom:** Phase transition coexistence curve for water near the triple point (from Mathez & Smerdon).

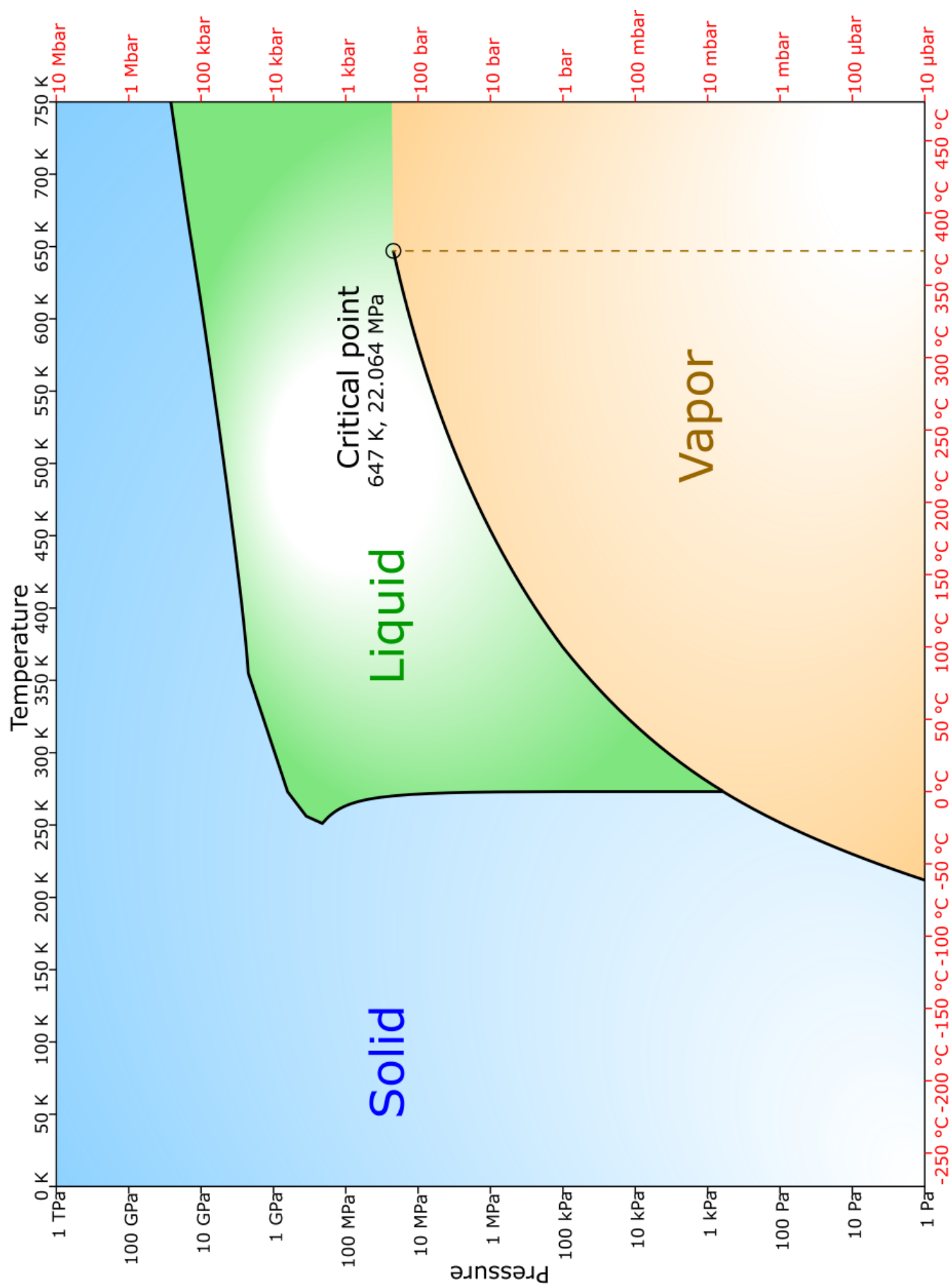


Figure 2: Phase diagram of water. (Adapted from wikimedia upload by Hokanomono and Cmglee)

The atmosphere and the radiative-convective equilibrium

The layers of the atmosphere, increasing from ground level (height, $h = 0$ km) are

- **Troposphere** [0–10 km]: Region of weather, contains 80% of total air mass.
 - **Stratosphere** [10–50 km] Ozone region where near-UV solar radiation is absorbed; temperatures increase to 0°C
 - **Mesosphere** [50–85 km] Low density of air (and ozone) causes temperature decrease to -100°C .
 - **Thermosphere** [> 85 km] Ionization of molecules by far-UV solar radiation causes temperatures to reach $500\text{--}2000^\circ\text{C}$ but pressure and density is extremely low (the “ionosphere” is here).
1. Let's draw a graph of height vs. temperature for the lower levels of the atmosphere. Take a regular sheet of graph paper in portrait orientation. Let the vertical axis go between 0 and 100 km, and the horizontal between -60 and 10°C .

- (a) Put a dot at $h = 0$ and $T = 15^\circ\text{C}$ (this will be just off the end of the lattice of the graph paper). This represents the Earth's surface at average temperature.
- (b) It can be shown with relatively simple physics (undergraduate thermodynamics) that warm rising air (convection) will cool in such a way that its temperature depends linearly on height:

$$\frac{\Delta T}{\Delta h} = -\Gamma$$

where Γ is called the *adiabatic lapse rate*. It's value for dry air (with zero humidity) can be shown to be almost exactly 10 K/km (or, equivalently $10\frac{^\circ\text{C}}{\text{km}}$), but the average value on Earth for *moist* air is about 6.5 K/km . Use this value to determine the temperature at the top of the troposphere (this location is called the *tropopause*).. Draw a dot there and use a ruler to draw the straight line decrease of temperature from surface to tropopause.

- (c) Sketch the temperature above 10 km using the information given above about the atmospheric layers.
- (d) The atmospheric pressure decreases exponentially with height:

$$p(h) = p_0 \exp\left[-\frac{h}{h_0}\right]$$

where $p_0 = 1\text{ atm}$ and the *scale height*, h_0 , is about 8 km . We can solve the above equation for height:

$$h = h_0 \ln\left[\frac{p_0}{p}\right]$$

use this to make an axis of pressure at the right of your graph paper. Give values of pressure on this axis at the corresponding heights where $p = \left[1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}\right] \times p_0$.

2. **Radiative-Convective Equilibrium Temperature** In prior weeks we made estimates of the Earth's equilibrium temperature. In fact this is the driving question of the first half of the course: What do we expect to be the temperature of the Earth? Let's review our prior results and then push forward to a new (and basically final) answer using the mechanism of energy transfer by evaporation.

- (a) Together with your group, remind yourselves of the two calculations we did previously for estimating the Earth's surface temperature: (i) Assuming no atmosphere, only the incoming solar flux ($S = 1360 \text{ J/s/m}^2$) and the average albedo of $\alpha = 0.3$; (ii) With an atmosphere that absorbs and then re-emits all of the Earth's outgoing (infrared) flux. What were the temperatures found in each case, and how do they compare to the observed average temperature of $\sim 15^\circ\text{C}$?
- (b) The *lapse rate* mentioned implies an additional flux of heat from the Earth's surface to the atmosphere. Redo the energy flow (or flux) diagram (e.g., Fig 4.5 of our textbook), but now include an additional arrow for "convective flux," F_c , from the Earth to the atmosphere.
- (c) Write down three energy balance equations ($\text{Flux}_{\text{in}} = \text{Flux}_{\text{out}}$) related to your figure from (b): (i) for the object of the whole Earth, (ii) for the atmosphere, and (iii) for the Earth's surface.
- (d) It turns out we do not need to "solve" these equations. We can use the *adiabatic lapse rate* together with your first equation (i) to find the temperature of the Earth's surface. Assume that the height of this atmosphere "object" is where the atmospheric pressure drops to half its surface value. What surface temperature do you obtain?

Milankovitch Cycles

1. The **Milankovitch cycles** are periodic oscillations of solar irradiance — recall that the “**Solar Constant**” is $S = 1360 \text{ J/s/m}^2$; we are talking about *variations of this number* over time — due to changes in the Earth’s **revolution** around the Sun and **rotation** on its axis.

(a) The cycles are due to three primary effects:

- Changes in the **eccentricity** of the Earth’s orbit around the Sun.
- Changes in the **obliquity** of the Earth’s rotational axis.
- Changes due to the **precession** of the Earth’s rotational axis.

Let’s identify in your own words what these things mean. Consider the information in: the class slides this week; Section 7.4 of our Dessler textbook; the brief NASA explanation and videos linked¹ on Blackboard (under Class Slides); the beginning of Chapter 7 of the Mathez textbook (also linked on Blackboard. Then:

- (i) Define these terms using your own words and pictures: eccentricity, obliquity, and precession.
 - (ii) Describe the three changes in these quantities (or changes associated to these quantities) as the Earth revolves and rotates.
 - (iii) Explain how each type of change would alter the solar irradiation of Earth (e.g., how would it change the timing and length of seasons in one hemisphere or another).
- (b) From the resources above, identify the **period** associated to each of the three Milankovitch cycle changes (i.e., the amount of time for each to complete one full cycle). Then, using graph paper (landscape orientation), draw three rows of graphs, where the horizontal axis is time and the scale is set to be the same for each. Choose that horizontal scale so that you can see at least two cycles of each (so at least *twice the longest period*). Then draw a sinusoidal curve² for each type of oscillation such that it matches the periods that you identified.
- (c) Let’s consider a “toy model” of oscillations for a moment. Imagine that the three different oscillations have periods of two, three, and five years respectively. If they were all to start at their peak values on the sinusoid (e.g., the orbital configuration representing maximal solar irradiance), after how much time would they arrive back at the same point, all at their peak values again simultaneously? Explain your answer in words, equations, and or pictures.

[**Bonus:** What is the general way that one can solve this problem, and is there always a solution? What is the answer for the Milankovitch cycles, i.e., how much time for all three types of cycles to return to their maximum values together?]

¹The NASA page is: <https://science.nasa.gov/science-research/earth-science/milankovitch-orbital-cycles-and-their-role-in-earths-climate/>.

²You will see in your readings and in the graphs below, that these periods are approximate. None of these effects are exactly described by a sinusoidal curve with a fixed period.

2. **Paleotemperature records** Figure ?? shows the Milankovitch oscillations for the past 800,000 years and coming 800,00 years.

- (a) Using a ruler, the timescale shown on the horizontal axis of Figure ??, and the information given in the figure caption, find the three periods you identified from question 1. For example, use a bar to indicate a time range containing an integer number (5–10?) of each cycle.
- (b) The prior time period — the past million years — is called the “ice ages” because of the cycles of “glacial” and “interglacial” periods. Identify on the figure our current interglacial period, and all prior glacial/interglacial periods.
- (c) It is said that this span of time is characterized by a 100 ky periodicity. Do you see any evidence for that in the temperature (proxy) record? Explain. What might cause the 100 ky periodicity? Do you see any connection between the Milankovitch cycles and the periodicity in the temperature record?
- (d) In Figure ?? we see a more zoomed in display of the past 400,000 years, together with the Milankovitch summary result (insolation at 65N latitude), taken from the original 1999 Nature paper that reported the Vostok ice core results. The authors have made some links between the temperature proxy data and the Milankovitch curve. Looking at this figure and reading through the original 1999 paper (available on blackboard in the Class Slides section), comment on these connections. What do the authors claim to be the connection here?

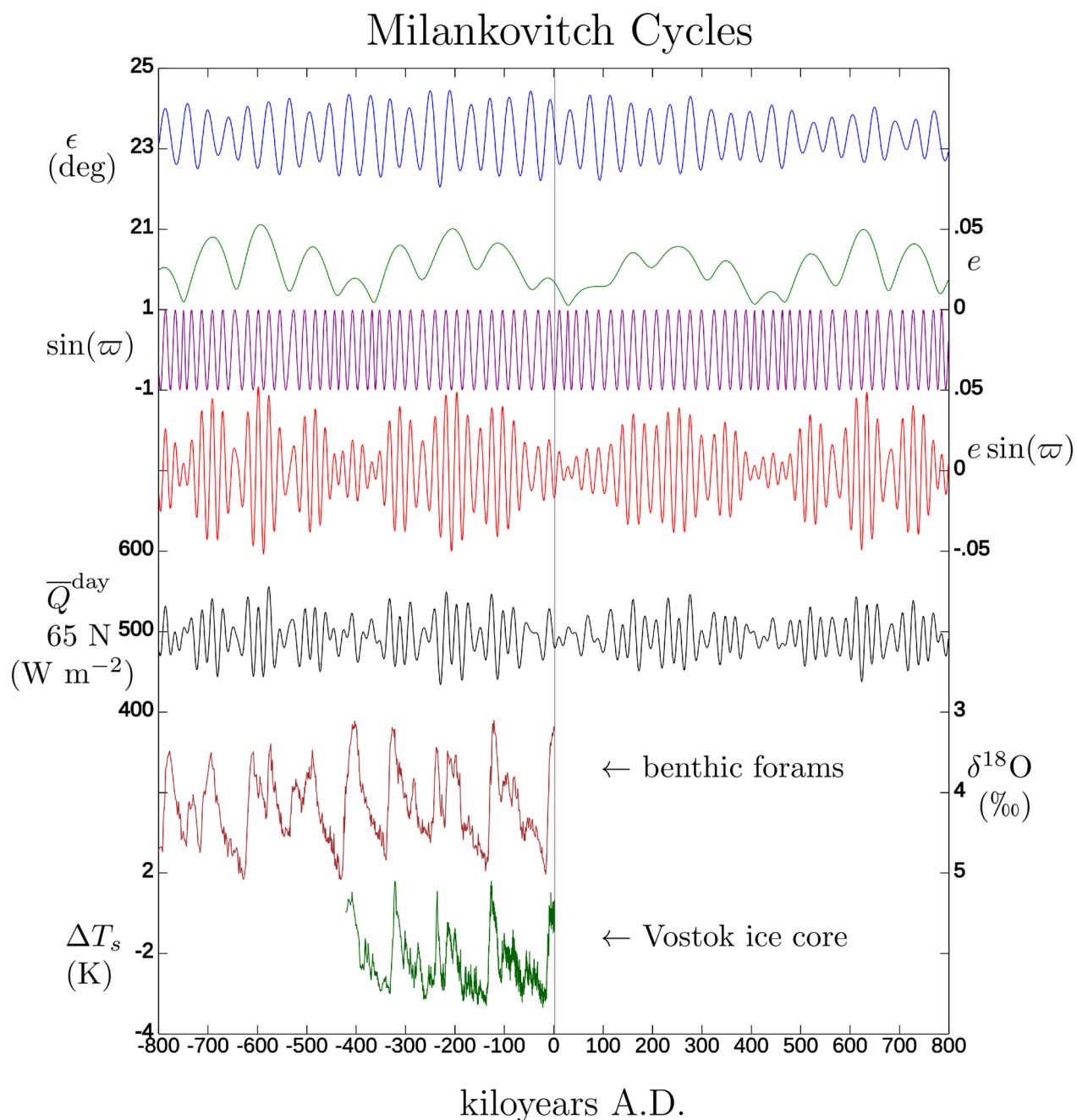


Figure 1: **First row:** Obliquity angle, ϵ , over time. **Second row:** Eccentricity, e , over time. **Third row:** Axis precession, $\sin \varpi$, over time. **Fourth row:** The “precession index” (eccentricity times precession), $e \sin \varpi$, over time. **Fifth row:** Estimated solar radiation at 65N latitude — found by multiplying the effect of obliquity and the precession index. **Sixth row:** Paleotemperature proxy from the incorporation of oxygen isotopes in shells of marine organisms. **Seventh row:** Paleotemperature (anomaly) estimates from Antarctic ice cores, from direct measurements of oxygen isotopes in trapped gas. [From Wikipedia by user incredio]

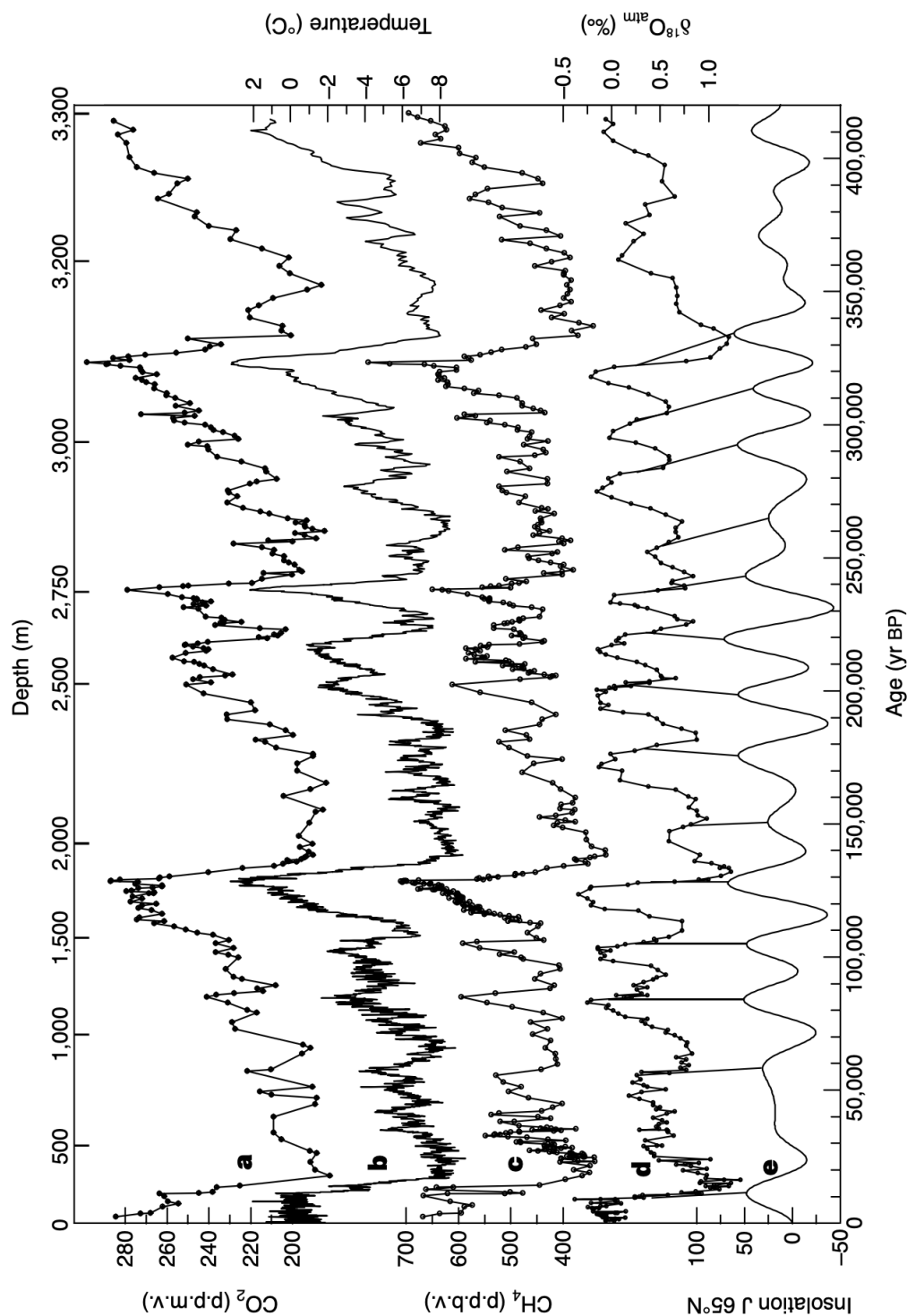


Figure 3 Vostok time series and insolation. Series with respect to time (GT4 timescale for ice on the lower axis, with indication of corresponding depths on the top axis) of: **a**, CO₂; **b**, isotopic temperature of the atmosphere (see text); **c**, CH₄; **d**, $\delta^{18}\text{O}_{\text{atm}}$; and **e**, mid-June insolation at 65° N (in W m^{-2}) (ref. 3). CO₂ and CH₄ measurements have been performed using the methods and analytical procedures previously described^{5,9}. However, the CO₂ measuring system has been slightly modified in order to increase the sensitivity of the CO₂ detection. The thermal conductivity chromatographic detector has been replaced by a flame ionization detector which measures CO₂ after its transformation into CH₄. The mean resolution of the CO₂ (CH₄) profile is about 1,500 (950) years. It goes up to about 6,000 years for CO₂ in the fractured zones and in the bottom part of the record, whereas the CH₄ time resolution ranges between a few tens of years to 4,500 years. The overall accuracy for CH₄ and CO₂ measurements are ± 20 p.p.b.v. and 2–3 p.p.m.v., respectively. No gravitational correction has been applied.

Figure 2: [Figure 3 from Petit et. al. (Nature 1999)]

Climate Sensitivity

The term **climate sensitivity** is used both as a technical term meaning *the Earth's equilibrium temperature increase due to a doubling in carbon dioxide concentration*¹, and as a general term to mean *the effect of increasing greenhouse gas concentrations on temperature*. Today we will look closely at the blackbody curve of the Earth's infrared emission (due to its average surface temperature of approximately $15^{\circ}\text{C} = 288\text{ K}$), and the CO_2 absorption lines in that spectrum, to understand (i) how greenhouse gas absorption causes temperature increase, and (ii) what we should expect that relationship to look like.

1. Let's start by carefully examining simulated blackbody curves of the Earth surface's emission to space as it would be² observed from space.

- (a) The blackbody curves shown in Figure 1 are plotted as a function of wavenumber. Convert our familiar wavelength for the carbon dioxide bending transition ($\lambda = 15\text{ }\mu\text{m}$), which takes the biggest “bite” out of the Earth's infrared blackbody spectrum, into wavenumber and frequency. Use the following equations to do so, expressing your final result in units of cm^{-1} for wavenumber and in units of $\text{THz} = 10^{12}\text{ Hz}$ for frequency:

$$f \lambda = c \quad \text{and} \quad \tilde{\nu} = \frac{1}{\lambda}$$

where $c = 3 \times 10^8\text{ m/s}$ is the speed of light, f is the variable for frequency, $\tilde{\nu}$ (“nu tilda”) is the variable most often used for wavenumber. Does your answer for wavenumber make sense based on the absorption lines seen in Figure 1? Does your answer for frequency make sense based on the graphs shown in Figure 3? Why or why not?

- (b) Looking at the graph in the top of Figure 1, what do you estimate to be the temperature of the emitted light? [In the regions where there are no bites, which blackbody curve would best fit?]
- (c) Why does the CO_2 absorption line have a flat base?
- (d) Why does the Ozone absorption line have a little peak in the middle? (Notice in the actual data plotted in Figure 1 (bottom) that the carbon dioxide also has such a peak.)
- (e) Approximate the $T = 240\text{ K}$ blackbody curve with a triangle by drawing it over the graph (the base of the triangle should lie on the horizontal axis). Calculate the area of this triangle using the values and units shown on the axes. What units does it have? What do we call this quantity? Does the number you found make sense based on our prior work?

¹I will use the phrase *carbon dioxide concentration* to mean the mole fraction of carbon dioxide in the air, which is just the fraction of all molecules in the air — regardless of the molecule type, size, or mass — that are CO_2 . This value is presently $425\text{ ppm} = \frac{425}{1000000} = 0.000425 = 0.0425\%$, meaning that 425 out of every million molecules of air are carbon dioxide. Nitrogen and Oxygen are the primary atmospheric gases with concentrations of 80% and 20%, respectively, so almost 800,000 and 200,000 per million particles are those. Methane, another greenhouse gas, has a current concentration of 1930 ppb (parts per billion), so two out of every million molecules of air, or 0.00019%.

²In fact we do measure the Earth's spectrum from space, using instruments on NASA's “Aqua” satellite using the AIRS and CERES instruments.

2. Now let's try to understand how CO₂ absorption lines lead to an increased overall temperature of the Earth.
- (a) Figure 2 shows simulated blackbody curves for the Earth's infrared emission, with different amounts of atmospheric CO₂. Describe generally how the absorption line changes as carbon dioxide is added to the atmosphere.
 - (b) When does the CO₂ absorption line "saturate" and what does that mean?
 - (c) On graph paper, draw an "approximate" blackbody curve as a (dashed-line) isosceles triangle with the following characteristics: base points at 5 μm and 25 μm ; a peak of 23.8 W/m²/ μm . What is its area under the curve?
 - (d) Make a carbon dioxide "absorption line" in your curve centered at 15 μm . Give it a depth that shows "saturation" (not all the way down to zero) and try to give it a width so that it "eats" $\frac{1}{3}$ to $\frac{1}{2}$ of the original area. What is the area of your triangle with the absorption line removed?
 - (e) Try to draw another isosceles triangle (with same base) that also has an absorption line of the same width and (fractional depth), but has total area equal to the original (un-eaten) triangle.

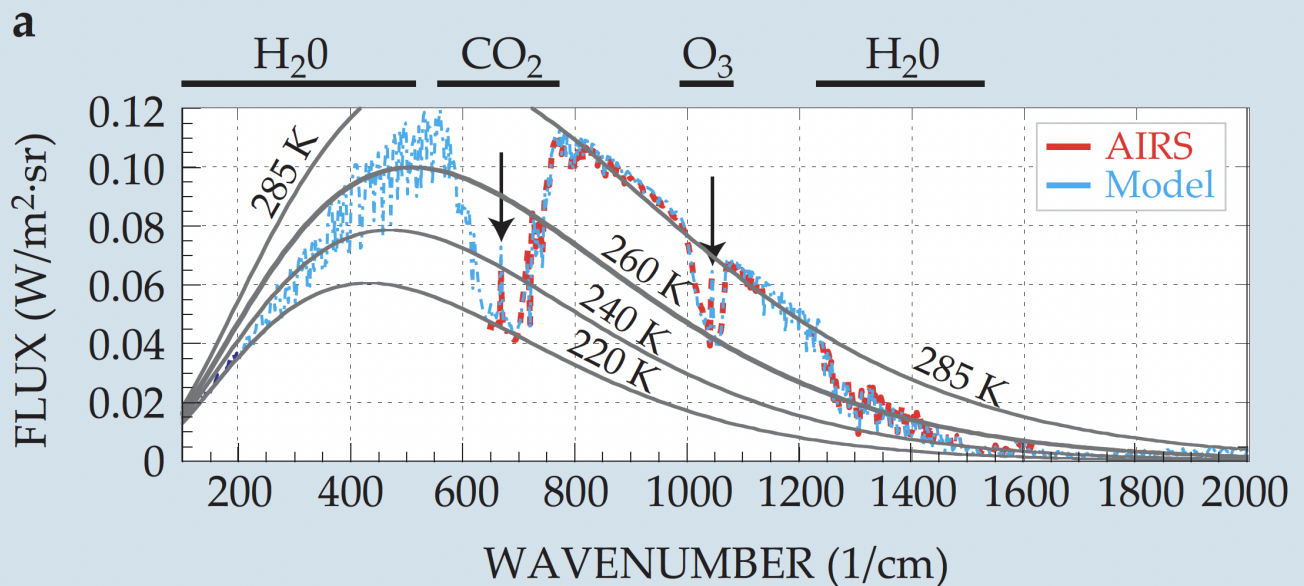
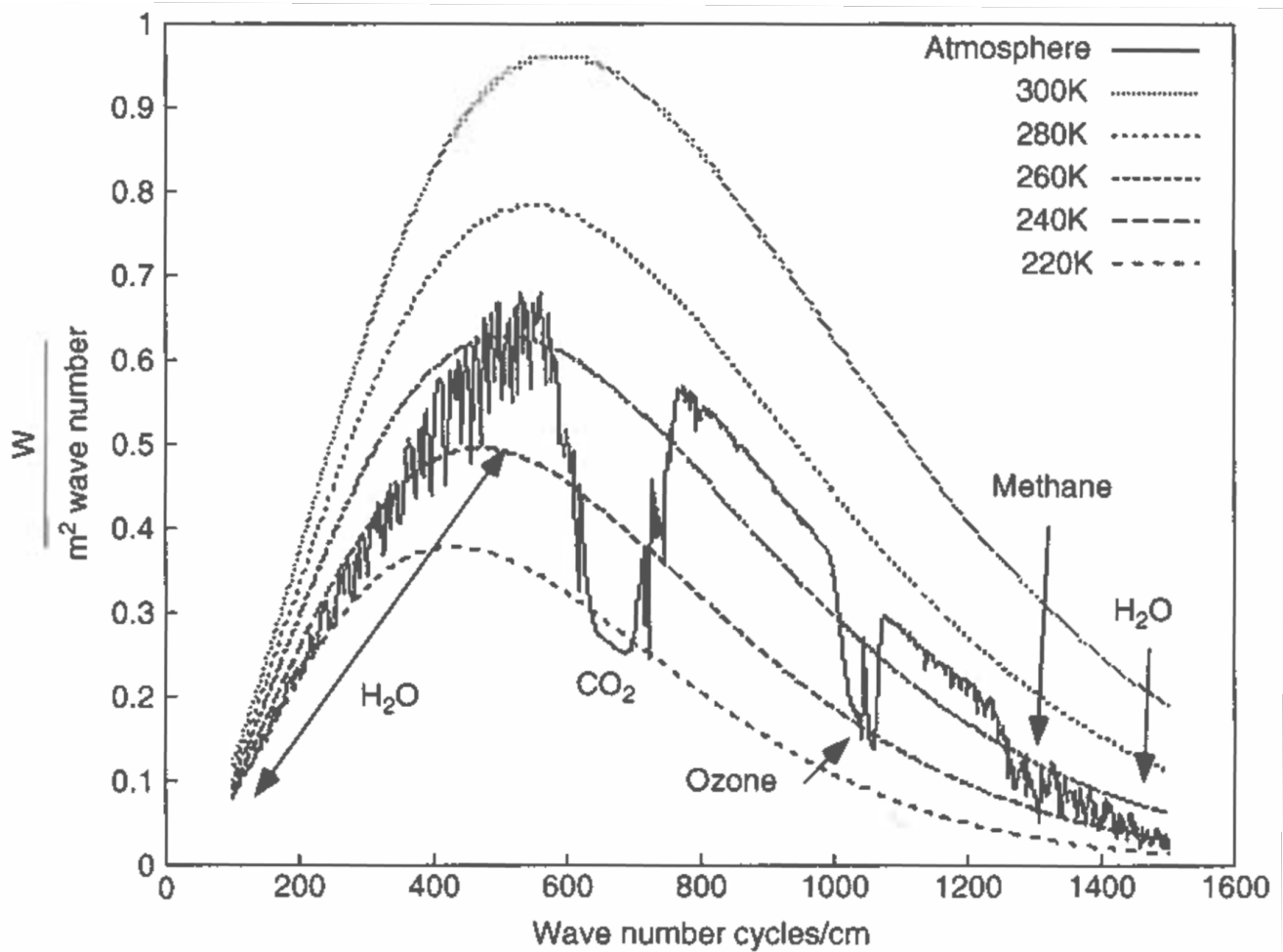


Figure 1: **Top:** Blackbody curves for different temperatures and a simulated spectrum of Earth as seen from space. [From Archer's *Global Warming: Understanding the Forecast*, Fig 4.3] **Bottom:** Similar graph including actual data from the NASA-AIRS satellite (AIRS) together with model simulated data (Model). [From Pierrehumbert's *Physics Today* article (January 2011)]

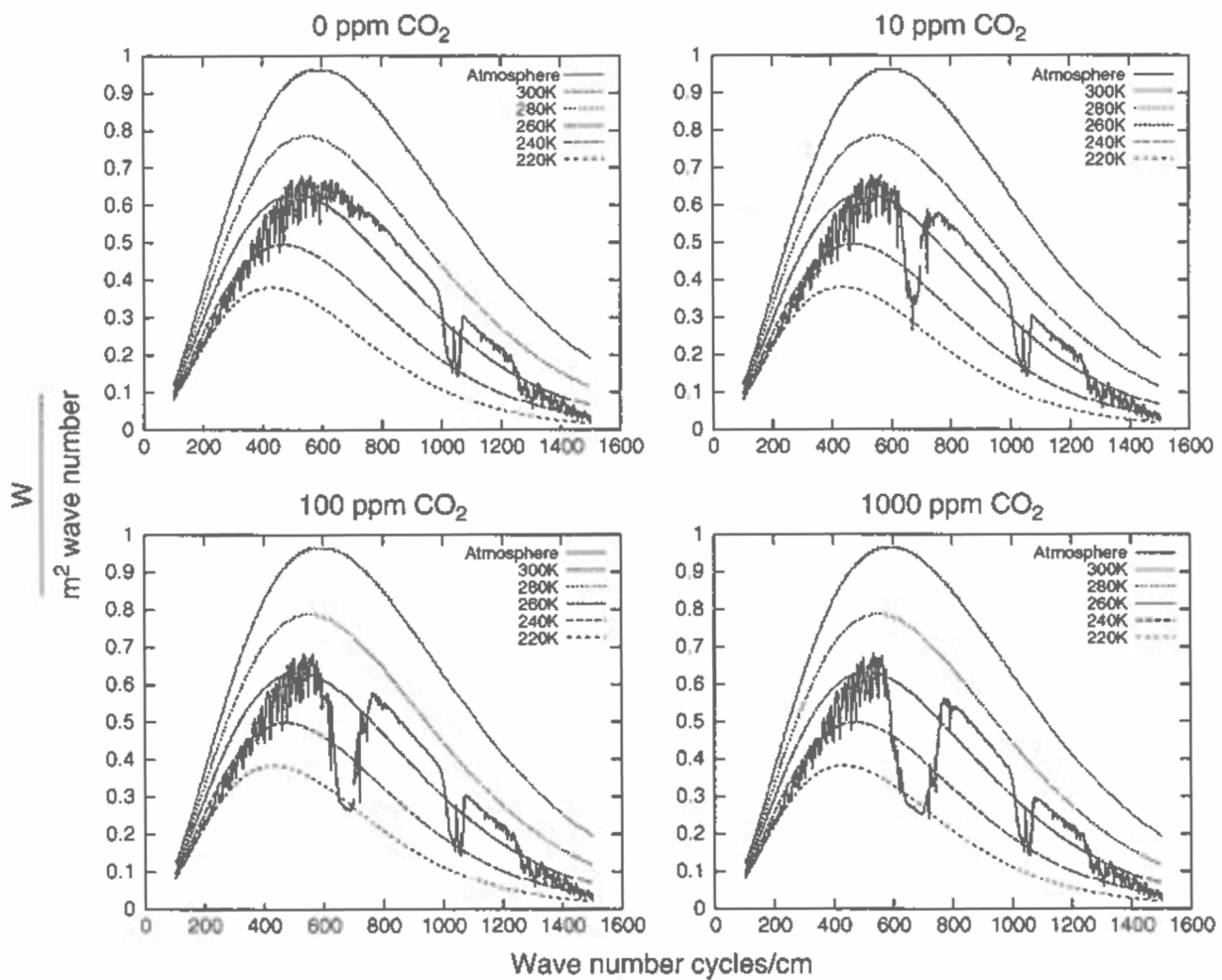


Figure 2: Blackbody curves for different temperatures and a simulated spectrum of Earth as seen from space, with different simulated amounts of atmospheric carbon dioxide. [From Archer's *Global Warming: Understanding the Forecast*, Fig 4.5]

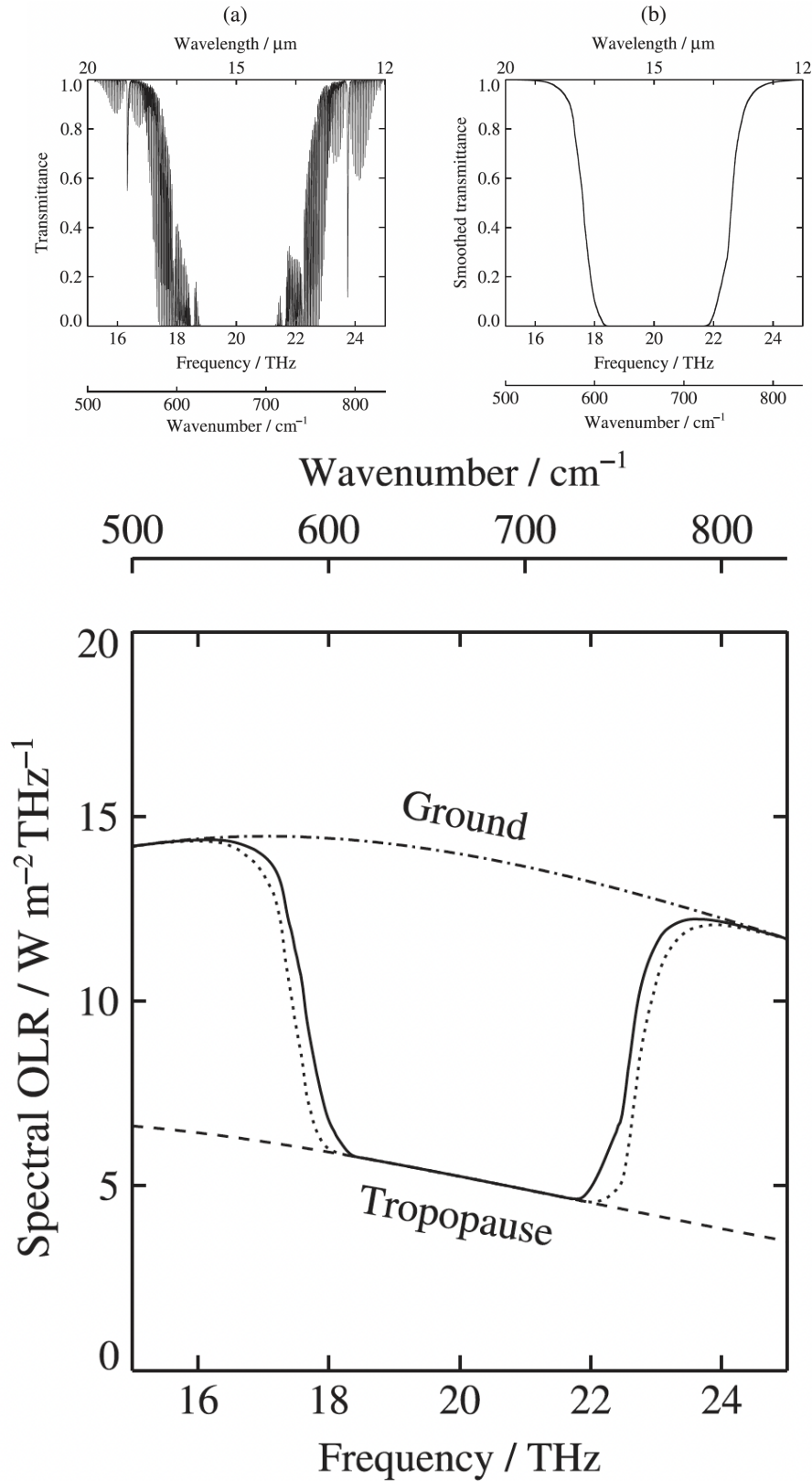


Figure 3: **Top:** Absorption spectrum for carbon dioxide, including actual quantum mechanical calculations (left) and smoothed approximation (right). The fine structure seen at left comes from rotational transitions on top of the primary bending transition at $15 \mu\text{m}$. [From Andrews' *Introduction to Atmospheric Physics* Fig 8.5] **Bottom:** Smoothed carbon dioxide bending transition superimposed on blackbody curve of Earth surface ($T = 290 \text{ K}$); the blackbody curve of the top of the troposphere is also shown ($T = 225 \text{ K}$). [From Andrews, Fig 8.7]

Climate Forcing and Timescales

1. Forcing, non-equilibrium, and return to equilibrium

- Using the information from this week's reading (*Dessler*, Chapter 6) and lecture, give a precise definition of **radiative forcing** (RF). Recall that it is a difference of two fluxes. What are its units?
- Explain the *meaning* of radiative forcing in the context of climate (radiative) equilibrium. Make reference to our prior work using energy flow/flux diagrams, and in particular the relationship of forcing to *equilibrium*. Explain the sequence of energy flux diagrams in *Dessler*'s Figure 6.1 (reproduced here in Figure 1). What is the value (and units) of the radiative forcing imposed by the sudden atmosphere?
- Repeat the calculation example done by *Dessler* in Section 6.3.1. Assume that the solar flux initially has its standard value¹ of 238 W/m^2 . Then, because of an orbital event, the Sun's flux increases by 1 W/m^2 , i.e., there is a radiative forcing of $+1 \text{ W/m}^2$. If the Earth's original equilibrium temperature (assuming no atmosphere) was 254.55 K , what will be the new temperature after it returns to equilibrium?
- The radiative forcing induced by all of the human carbon dioxide production since 1750 (changing the atmospheric concentration from 280 ppm to 425 ppm) is about $+2.2 \text{ W/m}^2$. The **climate sensitivity** — the forcing induced by a doubling of atmospheric carbon dioxide — is about $+4.0 \text{ W/m}^2$. Using your answer to part (c), which gave an estimate of the temperature increase per 1 W/m^2 of forcing, what are the corresponding temperature increase² predicted by these forcings for a no-atmosphere Earth?

2. Timescales of atmospheric carbon dioxide and temperature increase

- The change in atmospheric carbon dioxide over the past 70 years (the “Keeling Curve”) is shown in Figure 2. Let's try to characterize that data by plotting it ourselves.
 - First make a table with columns $[t(\text{yr}), C(\text{ppm}), \log_{10}(C)]$. Then choose about 5 points from the Keeling Curve data and record the values in your table, also calculating the logarithm of the concentration, C , value in your calculator.
 - Take a regular sheet of graph paper and plot $\log_{10}(C)$ vs time. Be sure to choose your axis ranges so that you can fill most of the paper.
 - Draw a best fit line to your plotted data and find its slope.

When a quantity is increasing exponentially, it can be seen as a straight line on a log vs linear plot. Does this data show evidence of exponential growth?

- For an exponentially growing quantity, we can assign a constant “doubling time” to the growth rate. This value is found as

$$t_{\text{double}} = \frac{\log_{10}(2)}{\text{slope}}$$

¹Remember that this value is $F_{\text{sun}} = \frac{S}{4}(1 - \alpha)$, where $S = 1360 \text{ J/s/m}^2$ is the Solar Constant, and $\alpha = 0.3$ is the albedo.

²Feedbacks in the climate system (e.g., ice melt $\rightarrow$ decreasing albedo $\rightarrow$ increasing F_{sun} , see *Dessler* Section 6.3.2) amplify the climate sensitivity. Current models incorporating all known effects estimate climate sensitivity to be about $3 \text{ K} = 3^\circ\text{C}$, with a likely range ($\sim 70\%$ probability of being in) 2.6 to 4°C .

where “slope” is the slope of the log-linear plot. Using this equation and your result from part (a), find the doubling time for atmospheric carbon dioxide.

- (c) Considering a longer-time plot of carbon dioxide in the atmosphere (Figure 3) and assuming that the exponential growth rate held over this whole period. When do you expect we will double the value of carbon dioxide from 1750?

3. (**Not used in F2025**) Consider the graphs in Figure 4. The top graph shows the effect of an annual 1% increase in CO₂ up to doubling. The bottom shows that the temperature is, model-predicted, to be proportional to the cumulative amount of emissions.

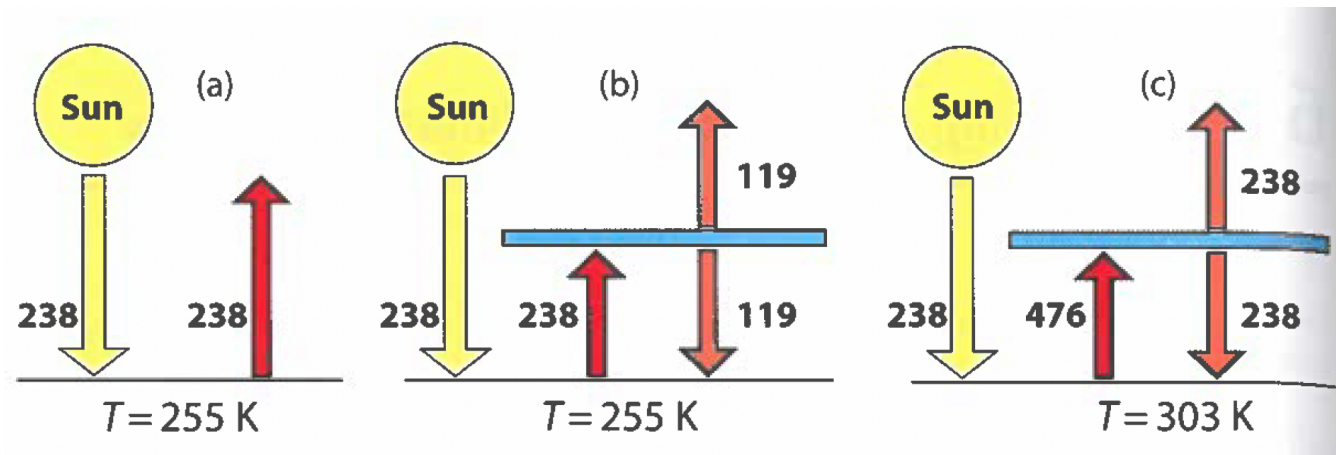


Figure 1: Figure 6.1 from *Dessler*: (a) An energy flux diagram for an Earth with no atmosphere (in equilibrium). (b) The immediate imposition of an atmosphere which is able to capture the Earth surface's IR radiation and re-radiate it. (c) A return to equilibrium.

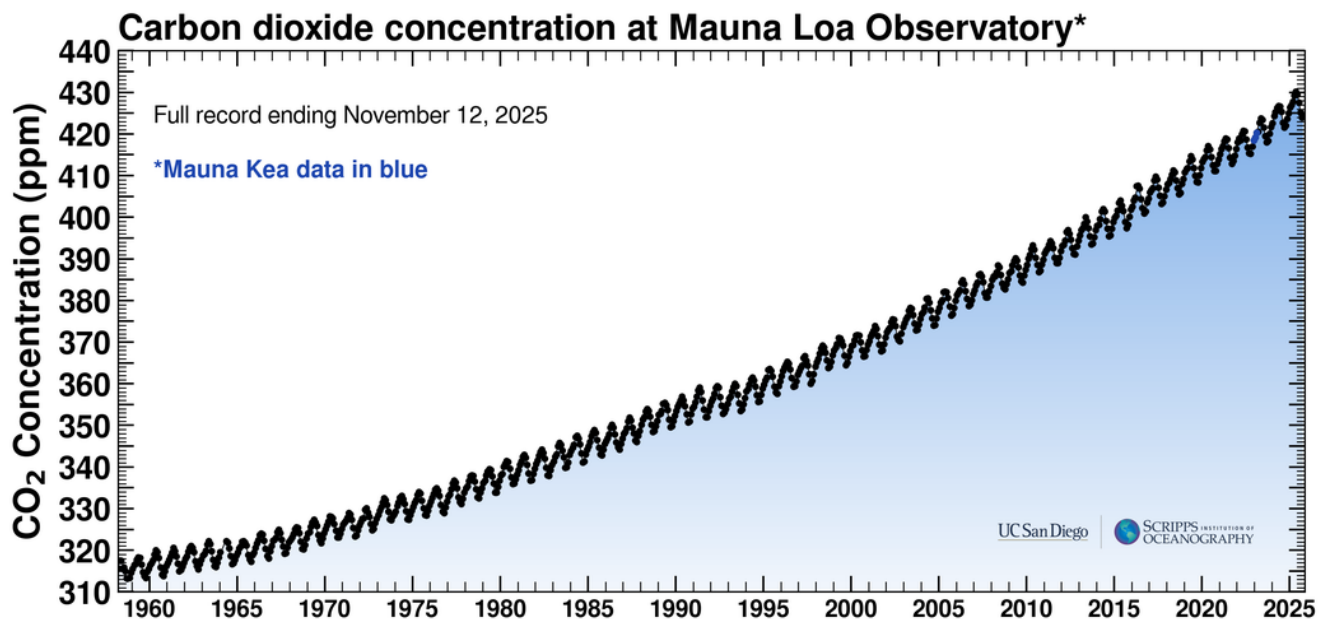


Figure 2: The so-called **Keeling Curve** (named after Charles Keeling, who began atmospheric CO₂ measurements at Mauna Loa Observatory in Hawaii, in 1956) showing atmospheric carbon dioxide as a function of time.

Global carbon dioxide emissions and atmospheric carbon dioxide (1751-2024)

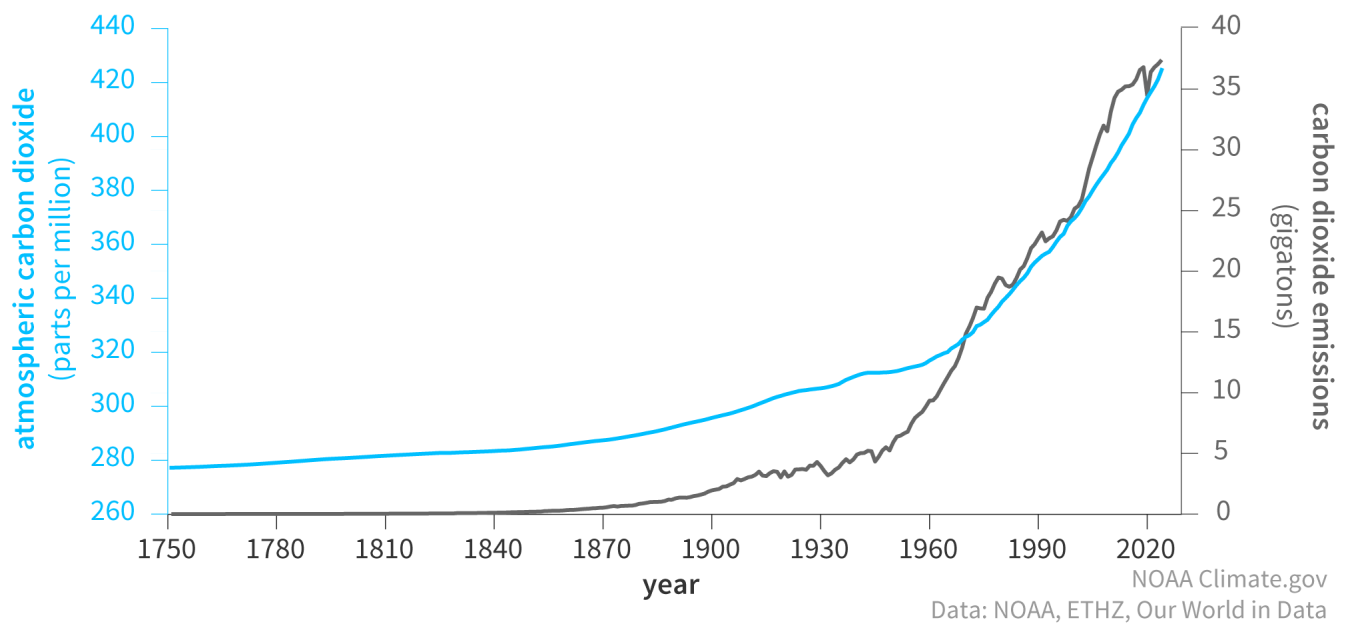


Figure 3: Atmospheric carbon dioxide and emissions since the Industrial Revolution.

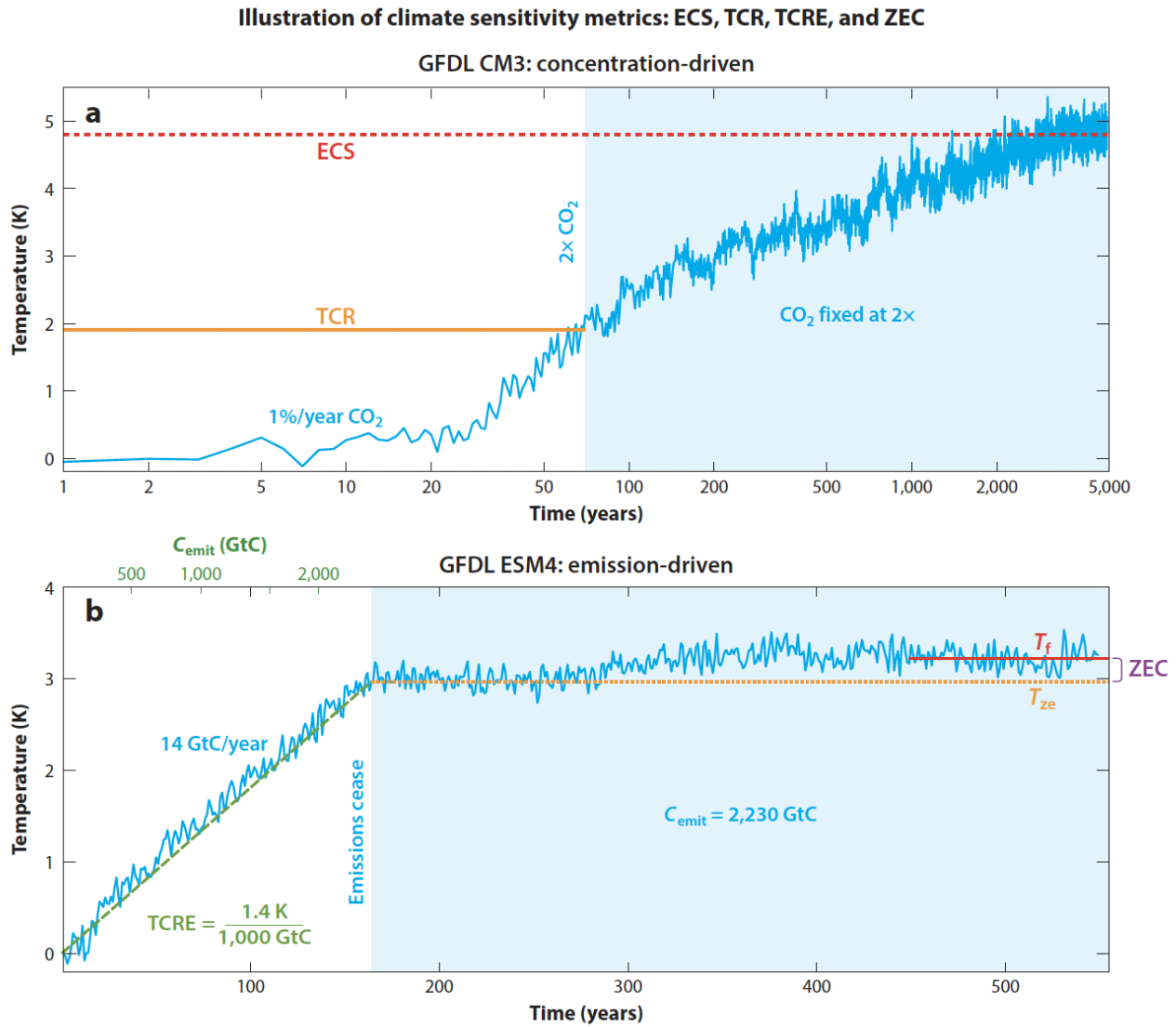


Figure 1

(a) Global mean surface temperature T_s versus time for a multi-millennial atmosphere-ocean GCM simulation using GFDL CM3 (Donner et al. 2011, Paynter et al. 2018). CO₂ concentrations increase at 1%/year until doubling at year 70 and are held fixed thereafter. TCR is the warming at year 70, and ECS is the equilibrium warming after several millennia. Note the log time axis.

(b) T_s versus time for an Earth system simulation with an active carbon cycle using GFDL ESM4. Constant emissions of roughly 14 GtC/year are prescribed until roughly 3°C of warming, at which point emissions cease. The TCRE is the slope of T_s versus cumulative emissions C_{emit} (green axis, upper left) while emissions are nonzero, and ZEC is the difference between the final warming T_f and the warming when emissions are zeroed T_{ze} . Abbreviations: ECS, equilibrium climate sensitivity; GFDL, Geophysical Fluid Dynamics Laboratory; GtC, gigatons carbon; TCR, transient climate response; TCRE, transient climate response to cumulative emissions; ZEC, zero emissions commitment.

Figure 4: Two pathways to the future.